# Numerical analysis of superconducting phases in the extended Hubbard model with non-local pairing

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**Abstract**

[To be continued...]

- Appendix on $s$-wave SC ghost simulations on HM
- Set ComputerModern font in all CairoMakie figures

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**List of symbols and abbreviations**

| | |
|---|---|
| AF | Antiferromagnetic |
| BCS | Bardeen-Cooper-Schrieffer (theory) |
| BZ | Brillouin zone |
| $\beta$ | Inverse temperature |
| $\Delta^{(\gamma)}$ | $\gamma$-wave component of the gap function |
| DoF | Degree of freedom |
| EHM | Extended Hubbard model |
| EV(s) | Expectation value(s) |
| FS | Fermi surface |
| HF | Hartree-Fock |
| HFP(s) | Hartree-Fock parameter(s) |
| HM | Hubbard model |
| $L_\ell$ | Number of lattice sites in direction $\ell$ |
| iHF | Iterative Hartree-Fock |
| LRT | Linear Response Theory |
| MBZ | Magnetic Brillouin zone |
| MFT | Mean-Field Theory |
| NBZ | Nematic Brillouin zone |
| NN(s) | Nearest neighbor(s) |
| PH | Particle-hole |
| PHS | Particle-hole symmetry |
| RWC(s) | Relevant Wick's contraction(s) |
| RMP(s) | Renormalized model parameter(s) |
| SC | Superconductor |
| SDW | Spin-density wave |
| SSB | Spontaneous symmetry breaking |
| $T_\mathrm{c}$ | Critical temperature |

# Introduction

This thesis project is about my favorite ice cream flavor. [To be continued...]

# Chapter 1

# Superconducting instability

In this chapter the HF methods introduced in Sec. **??** are applied to the SC phase. We study a translationally invariant superconductor, which breaks the conservation of total charge. Analytic calculations will lead us to various effects: bare bands renormalisation, anisotropic Cooper pairing and nematicity. Our description of the SC phase is closely related to the BCS solution for pure $s$-wave superconductivity.

As for previous phases, this chapter is divided in three parts. First, analysis of the SC phase symmetries based on the discussion of Chap. **??**, and implementation of HF methods. Second, search for theoretical constraints on the self-consistent solution. Third, numeric results.

## 1.1 The SC phase and its symmetries

As for antiferromagnetism, superconductivity can establish in a variety of ways. We deal with the simplest realisation. In Sec. **??** we anticipated that the SC phase is described by the two SSBs

$$\mathrm{U}^z(1) \overset{\mathrm{SSB}}{\to} 0$$
$$\mathrm{C}_4 \overset{\mathrm{SSB}}{\to} \mathrm{C}_2$$

Translational invariance is unbroken. Then electronic density needs to be uniform across all sites, similarly to the normal phase discussed in Chap. **??**. Non-uniform density distributions are inconsistent with translational invariance in both directions. We have, then

$$\langle \hat{n}_{\mathbf{r}\sigma} \rangle = n \tag{1.1}$$

which is identical to Eq. (**??**) for the normal phase.

Recall Tabs. **??** and **??**, collecting the selection rules for the local repulsion $\hat{H}_U$ and the non-local attraction $\hat{H}_V$. From Eqns. (1.2), (**??**) for the SC phase we need to account for the following contractions:

| | |
|---|---|
| $\hat{H}_U$ | Hartree and Cooper contractions |
| $\hat{H}_V$ (o.s. sector) | Hartree and Cooper contractions |
| $\hat{H}_V$ (s.s. sector) | Hartree and Fock contractions |

We anticipate their roles in Tab. 1.1.

The SSB of $\mathrm{U}^z(1)$ allows for EVs like $\langle \hat{c}^\dagger \hat{c}^\dagger \rangle$ to be non-zero. This leads to the formation of Cooper pairs. These are composite quantum objects with a specific spin-spatial structure. Consider the generic Cooper fluctuation

$$\left\langle \hat{c}^\dagger_{i\sigma} \hat{c}^\dagger_{j\bar{\sigma}} \right\rangle \quad \text{with } i,j \text{ nearest neighbours}$$

We work in o.s. sector because we know from Tab. **??** that the s.s. sector is prohibited. From basic Quantum Mechanics we know the summation rules of the spin algebra $\mathfrak{su}(2)$,

$$\frac{1}{2} \otimes \frac{1}{2} = 0 \oplus 1$$

| Operator | Sector | RWCs | Net effect |
|---|---|---|---|
| $\hat{H}_U$ | | Hartree | $\mu$ shift by $-nU$ |
| | | Cooper | Isotropic pairing in singlet channel |
| $\hat{H}_V$ | o.s. | Hartree | $\mu$ shift by $nzV$ |
| | | Cooper | Anisotropic pairing in singlet and triplet channels |
| | s.s. | Hartree | $\mu$ shift by $nzV$ |
| | | Fock | Band renormalisation and resymmetrisation |

Table 1.1 Relevant Wick's contractions and net effect in the SC phase. This RWCs table is more structured than Tabs. **??** and **??**.

| Spatial structure | Pairing channel |
|---|---|
| Symmetric wave function | Singlet pairing |
| Anti-symmetric wave function | Triplet pairing |

Table 1.2 Relation of the Cooper pairing channel with the wavefunction spatial symmetry (intended as the inversion $(x, y) \to (-x, -y)$).

The two pairing channels are, at this level, the singlet channel associated to total spin 0 and the triplet channel associated to total spin 1. Now, when applying HF method to the quartic operators, we get the decomposition of Eq. (**??**). We can impose to the approximated HF ground-state a particular spatial structure – say, inversion symmetry or antisymmetry. This reflects in the requirement that Cooper pairs show the same spatial structure.

This gives a selection rule over Cooper pairings: if we work with inversion-symmetric structures – say, $s^*$-wave or $d$-wave – the pairing will take place in singlet channel, allowing us to eliminate the triplet pairing. Working with antisymmetric structures, inversely, we can eliminate the singlet channel. This concept is summarised in Tab. 1.2.

We proceed as we did for the AF phase: in order to highlight the physical effects of $\hat{H}_V$, we first analyse the SC phase limitedly to the conventional HM. In next section we prove that BCS-like superconductivity is impossible at HF level on the pure HM, with just the local repulsion $\hat{H}_U$. It is the presence of the non-local attraction that enables Cooper pairing mechanism. Before starting, let

$$\hat{n}_{\mathbf{k}\sigma} \equiv \hat{c}^{\dagger}_{\mathbf{k}\sigma} \hat{c}_{\mathbf{k}\sigma} \tag{1.2}$$

$$\hat{\phi}_{\mathbf{k}} \equiv \hat{c}_{-\mathbf{k}\downarrow} \hat{c}_{\mathbf{k}\uparrow} \tag{1.3}$$

where $\hat{n}_{\mathbf{k}\sigma}$ was defined as for the AF phase, in Eq. (**??**). The HF approximation in reciprocal space to the EHM will be written in terms of these operators.

## 1.2 Local effects: (impossibility of) superconductivity in the HM

Let us start from the local effects. Following Tab. **??** and Eqns. (1.2), (**??**), we know the local repulsion decomposes in Hartree and Cooper channels,

$$\begin{aligned} \hat{H}_U &= U \sum_{\mathbf{r}\in\mathcal{L}} \hat{n}_{\mathbf{r}\uparrow} \hat{n}_{\mathbf{r}\downarrow} \\ &\simeq U \sum_{\mathbf{r}\in\mathcal{L}} \Big( \hat{c}^{\dagger}_{\mathbf{r}\uparrow}\hat{c}_{\mathbf{r}\uparrow} \langle \hat{c}^{\dagger}_{\mathbf{r}\downarrow}\hat{c}_{\mathbf{r}\downarrow} \rangle + \langle \hat{c}^{\dagger}_{\mathbf{r}\uparrow}\hat{c}_{\mathbf{r}\uparrow} \rangle \hat{c}^{\dagger}_{\mathbf{r}\downarrow}\hat{c}_{\mathbf{r}\downarrow} - \langle \hat{c}^{\dagger}_{\mathbf{r}\uparrow}\hat{c}_{\mathbf{r}\uparrow} \rangle \langle \hat{c}^{\dagger}_{\mathbf{r}\downarrow}\hat{c}_{\mathbf{r}\downarrow} \rangle && \text{(Hartree)} \\ &\qquad + \hat{c}^{\dagger}_{\mathbf{r}\uparrow}\hat{c}^{\dagger}_{\mathbf{r}\downarrow} \langle \hat{c}_{\mathbf{r}\downarrow}\hat{c}_{\mathbf{r}\uparrow} \rangle + \langle \hat{c}^{\dagger}_{\mathbf{r}\uparrow}\hat{c}^{\dagger}_{\mathbf{r}\downarrow} \rangle \hat{c}_{\mathbf{r}\downarrow}\hat{c}_{\mathbf{r}\uparrow} - \langle \hat{c}^{\dagger}_{\mathbf{r}\uparrow}\hat{c}^{\dagger}_{\mathbf{r}\downarrow} \rangle \langle \hat{c}_{\mathbf{r}\downarrow}\hat{c}_{\mathbf{r}\uparrow} \rangle \Big) && \text{(Cooper)} \end{aligned}$$

In next sections we deal with each channel separately.

### 1.2.1 Local Hartree channel in the SC phase: analytic derivation

For the Hartree channel, being the density uniform across sites, we can re-use the entirety of Sec. **??** for the normal phase. Substituting Eq. (1.1), we get the approximation of Eq. (**??**),

$$\hat{H}_U \simeq nU \times \hat{N} - E^{(\mathrm{SC})}_{\mathrm{H}/U} + (\text{Cooper channel}) \tag{1.4}$$

thus chemical potential renormalization $\tilde{\mu} \to \mu - nU$ is present also in the SC phase.

### 1.2.2 Local Hartree contraction energy

As for the section above, we use the result of Sec. **??** for the normal phase. From Eq. (**??**)

$$E^{(\mathrm{SC})}_{\mathrm{H}/U} = L_x L_y \times n^2 U \tag{1.5}$$

For a locally repulsive model, the Hartree channel contributes negatively to ground state energy. Indeed, contraction energy is subtracted from the hamiltonian. We now move to the Cooper channel.

### 1.2.3 Local Cooper channel in the SC phase: analytic derivation

We move to reciprocal space, through the transformation of Eq. (**??**). Consider the Cooper channel of Eq. (1.2). In reciprocal space we get:

$$\begin{aligned}\hat{H}_U \simeq (\text{Hartree channel}) + \frac{U}{L_x L_y} \sum_{\mathbf{K},\mathbf{k},\mathbf{k}'} \Big[ &\big\langle \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\uparrow} \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\downarrow} \big\rangle \hat{c}_{\mathbf{K}-\mathbf{k}'\downarrow} \hat{c}_{\mathbf{K}+\mathbf{k}'\uparrow} \\ &+ \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\uparrow} \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\downarrow} \big\langle \hat{c}_{\mathbf{K}-\mathbf{k}'\downarrow} \hat{c}_{\mathbf{K}+\mathbf{k}'\uparrow} \big\rangle - \big\langle \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\uparrow} \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\downarrow} \big\rangle \big\langle \hat{c}_{\mathbf{K}-\mathbf{k}'\downarrow} \hat{c}_{\mathbf{K}+\mathbf{k}'\uparrow} \big\rangle \Big]\end{aligned} \tag{1.6}$$

We are not breaking translational invariance, thus only Cooper fluctuations with net zero total momentum are allowed. This means only $\mathbf{K} = \mathbf{0}$ contributes. Let us prove this: consider the EV of the Cooper pair

$$\big\langle \hat{c}^\dagger_{\mathbf{k}_1\uparrow} \hat{c}^\dagger_{\mathbf{k}_2\downarrow} \big\rangle$$

for two momenta $\mathbf{k}_1$, $\mathbf{k}_2$. Moving to real space via an $\mathcal{F}$-transform, we get

$$\big\langle \hat{c}^\dagger_{\mathbf{k}_1\uparrow} \hat{c}^\dagger_{\mathbf{k}_2\downarrow} \big\rangle \stackrel{\mathcal{F}}{=} \frac{1}{L_x L_y} \sum_{\mathbf{r}_1 \in \mathcal{L}} \sum_{\mathbf{r}_2 \in \mathcal{L}} e^{-i(\mathbf{k}_1 \cdot \mathbf{r}_1 + \mathbf{k}_2 \cdot \mathbf{r}_2)} \big\langle \hat{c}^\dagger_{\mathbf{r}_1\uparrow} \hat{c}^\dagger_{\mathbf{r}_2\downarrow} \big\rangle$$

where we used the convention of Eq. (**??**). Let now $\mathbf{r}_1 \to \mathbf{r}$ and $\mathbf{r}_2 \to \Delta\mathbf{r}$. Translational invariance implies that

$$g(\Delta\mathbf{r}) = \big\langle \hat{c}^\dagger_{\mathbf{r}\uparrow} \hat{c}^\dagger_{\mathbf{r}+\Delta\mathbf{r}\downarrow} \big\rangle$$

is independent of $\mathbf{r}$. Thus we have:

$$\begin{aligned}\big\langle \hat{c}^\dagger_{\mathbf{k}_1\uparrow} \hat{c}^\dagger_{\mathbf{k}_2\downarrow} \big\rangle &\stackrel{\mathcal{F}}{=} \sum_{\Delta\mathbf{r} \in \mathcal{L}} e^{-i\mathbf{k}_2 \cdot \Delta\mathbf{r}} g(\Delta\mathbf{r}) \frac{1}{L_x L_y} \sum_{\mathbf{r} \in \mathcal{L}} e^{-i(\mathbf{k}_1 + \mathbf{k}_2) \cdot \mathbf{r}} \\ &= \sum_{\Delta\mathbf{r} \in \mathcal{L}} e^{-i\mathbf{k}_2 \cdot \Delta\mathbf{r}} g(\Delta\mathbf{r}) \delta_{\mathbf{k}_1 = \mathbf{k}_2}\end{aligned}$$

where we used the identity of Eq. (**??**). Then the above EV vanishes for $\mathbf{k}_1 \neq \mathbf{k}_2$. Getting back to Eq. (1.6), this proves that only pairs at null total momentum can have nonzero EVs.

Let now $w_{\mathbf{k}}$ be the EV of $\phi^\dagger_{\mathbf{k}}$, and $w^{(s)}$ its $s$-wave projection:

$$w_{\mathbf{k}} \equiv \big\langle \phi^\dagger_{\mathbf{k}} \big\rangle \tag{1.7}$$

$$w^{(s)} \equiv \frac{1}{L_x L_y} \sum_{\mathbf{q} \in \mathrm{BZ}} w_{\mathbf{k}} \tag{1.8}$$

Eq. (1.6) reduces to

$$\hat{H}_U \simeq (\text{Hartree channel}) + \frac{U}{L_x L_y} \sum_{\mathbf{k},\mathbf{k}'} \left[ \langle \hat{\phi}^\dagger_{\mathbf{k}} \rangle \hat{\phi}_{\mathbf{k}'} + \hat{\phi}^\dagger_{\mathbf{k}} \langle \hat{\phi}_{\mathbf{k}'} \rangle \right] - E^{(\mathrm{SC})}_{\mathrm{C}/U} \tag{1.9}$$

being $E^{(\mathrm{SC})}_{\mathrm{C}/U}$ the HM Cooper contraction energy, written explicitly in Sec. 1.2.4. Let now

$$\Delta^{(s)} \equiv -U w^{(s)} \tag{1.10}$$

This is the $s$-wave component of the gap function. Note that, in general, $\Delta^{(s)} \in \mathbb{C}$. However, we can always choose $\Delta^{(s)}$ as purely real. This is due to the fact that the hamiltonian is gauge-invariant, then it suffices to apply the gauge map of Eq. (**??**) to the fermionic operators with

$$\eta \equiv \frac{\arg \Delta^{(s)}}{2}$$

to obtain $\Delta^{(s)} \in \mathbb{R}$. For the sake of generality, we proceed with $\Delta^{(s)} \in \mathbb{C}$. Although this choice gives no advantage now, we implement it anyway in order to derive the most general form of the equations that follow. This is useful for the generalisation to the EHM.

In the end, we reduce $\hat{H}_U$ from Eq. (1.9) to

$$\hat{H}_U \simeq (\text{Hartree channel}) - \sum_{\mathbf{k}\in\mathrm{BZ}} \left[ \Delta^{(s)} \hat{\phi}_{\mathbf{k}} + \Delta^{(s)*} \hat{\phi}^\dagger_{\mathbf{k}} \right] - E^{(\mathrm{SC})}_{\mathrm{C}/U} \tag{1.11}$$

This last decomposition resembles Eq. (**??**), obtained for the AF phase in the Hartree channel.

### 1.2.4 Local Cooper contraction energy

The Cooper contraction energy in the Cooper channel is given by:

$$\begin{aligned} E^{(\mathrm{SC})}_{\mathrm{C}/U} &\equiv \frac{U}{L_x L_y} \sum_{\mathbf{k}\mathbf{k}'} \langle \hat{\phi}^\dagger_{\mathbf{k}} \rangle \langle \hat{\phi}_{\mathbf{k}'} \rangle \\ &= L_x L_y \times U \left| w^{(s)} \right|^2 \end{aligned} \tag{1.12}$$

As for the Hartree channel, also the Cooper channel contributes negatively to ground-state energy.

### 1.2.5 Superconductivity in the HM

We proceed here as we did for the AF phase in Sec. **??**. We start by giving a matrix representation of the approximated hamiltonian, then map it to a pseudospin problem and finally specify how to diagonalise it to retrieve its ground state.

**Matrix representation.** Collecting the results of the above paragraphs, we obtain the quadratic approximation to the Hubbard hamiltonian in the SC phase,

$$\begin{aligned} \hat{H}_t + \hat{H}_U \simeq & \sum_{\mathbf{k}\in\mathrm{BZ}} \sum_{\sigma} \epsilon_{\mathbf{k}\sigma} \hat{c}^\dagger_{\mathbf{k}\sigma} \hat{c}_{\mathbf{k}\sigma} \\ & - \sum_{\mathbf{k}\in\mathrm{BZ}} \left[ \Delta^{(s)} \hat{\phi}_{\mathbf{k}} + \Delta^{(s)*} \hat{\phi}^\dagger_{\mathbf{k}} \right] + nU \times \hat{N} - \left[ E^{(\mathrm{SC})}_{\mathrm{H}/U} + E^{(\mathrm{SC})}_{\mathrm{C}/U} \right] \end{aligned} \tag{1.13}$$

We define the shifted bare bands:

$$\xi_{\mathbf{k}} \equiv \epsilon_{\mathbf{k}} - \tilde{\mu}$$

Notice we are using the renormalised chemical potential. Due to inversion symmetry, it holds $\xi_{\mathbf{k}} = \xi_{-\mathbf{k}}$. Using Fermi anticommutation rules, the kinetic operator can be written as:

$$\begin{aligned}\hat{H}_t - \tilde{\mu}\hat{N} &= \sum_{\sigma}\sum_{\mathbf{k}\in \mathrm{BZ}} \xi_{\mathbf{k}\sigma}\hat{n}_{\mathbf{k}\sigma} \\ &= \sum_{\mathbf{k}\in \mathrm{BZ}} \left(\xi_{\mathbf{k}}\hat{n}_{\mathbf{k}\uparrow} + \xi_{-\mathbf{k}}\hat{n}_{-\mathbf{k}\downarrow}\right) \\ &= \sum_{\mathbf{k}\in \mathrm{BZ}} \xi_{\mathbf{k}} \left(\hat{c}^{\dagger}_{\mathbf{k}\uparrow}\hat{c}_{\mathbf{k}\uparrow} - \hat{c}_{-\mathbf{k}\downarrow}\hat{c}^{\dagger}_{-\mathbf{k}\downarrow} + 1\right) \\ &= \sum_{\mathbf{k}\in \mathrm{BZ}} \xi_{\mathbf{k}} \left(\hat{c}^{\dagger}_{\mathbf{k}\uparrow}\hat{c}_{\mathbf{k}\uparrow} - \hat{c}_{-\mathbf{k}\downarrow}\hat{c}^{\dagger}_{-\mathbf{k}\downarrow}\right) + E^{(\mathrm{SC})}_{\mathrm{a}}\end{aligned}$$

being $E^{(\mathrm{SC})}_{\mathrm{a}}$ the “anticommutation energy”

$$E^{(\mathrm{SC})}_{\mathrm{a}} \equiv \sum_{\mathbf{k}\in \mathrm{BZ}} \xi_{\mathbf{k}} \tag{1.14}$$

In the end, the hamiltonian is approximated by:

$$\begin{aligned}\hat{H}_t + \hat{H}_U \simeq \sum_{\mathbf{k}\in \mathrm{BZ}} \epsilon_{\mathbf{k}} \left[\hat{c}^{\dagger}_{\mathbf{k}\uparrow}\hat{c}_{\mathbf{k}\uparrow} - \hat{c}_{\mathbf{k}\downarrow}\hat{c}^{\dagger}_{\mathbf{k}\downarrow}\right] + E^{(\mathrm{SC})}_{\mathrm{a}} \\ - \sum_{\mathbf{q}\in \mathrm{BZ}} \left[\Delta^{(s)}\hat{\phi}_{\mathbf{q}} + \Delta^{(s)*}\hat{\phi}^{\dagger}_{\mathbf{q}}\right] + nU \times \hat{N} - \left[E^{(\mathrm{SC})}_{\mathrm{H}/U} + E^{(\mathrm{SC})}_{\mathrm{C}/U}\right]\end{aligned}$$

with $E^{(\mathrm{SC})}_{\mathrm{H}/U}$, $E^{(\mathrm{SC})}_{\mathrm{C}/U}$ the contraction energies arising from Hartree and Cooper channels, and $E^{(\mathrm{SC})}_{\mathrm{a}}$ the energy shift resulting from anticommutation rules, given respectively in Eqns. (1.5), (1.12), and (1.14). We recognised the $s$-wave gap $\Delta^{(s)}$ from Eq. (1.10).

Let the Nambu spinor

$$\hat{\Phi}_{\mathbf{k}} \equiv \begin{bmatrix} \hat{c}_{\mathbf{k}\uparrow} \\ \hat{c}^{\dagger}_{-\mathbf{k}\downarrow} \end{bmatrix}$$

The approximated hamiltonian can be written in matrix form as:

$$\hat{H}_t + \hat{H}_U - \mu\hat{N} \simeq \sum_{\mathbf{k}\in \mathrm{BZ}}\sum_{\sigma} \hat{\Phi}^{\dagger}_{\mathbf{k}} h_{\mathbf{k}} \hat{\Phi}_{\mathbf{k}} + E^{(\mathrm{SC})}_{\mathrm{a}} - \left[E^{(\mathrm{SC})}_{\mathrm{H}/U} + E^{(\mathrm{SC})}_{\mathrm{C}/U}\right]$$

being

$$h_{\mathbf{k}} \equiv \begin{bmatrix} \xi_{\mathbf{k}} & -\Delta^{*}_{\mathbf{k}} \\ -\Delta_{\mathbf{k}} & -\xi_{\mathbf{k}} \end{bmatrix} \quad \text{with} \quad \xi_{\mathbf{k}} \equiv \epsilon_{\mathbf{k}} - \tilde{\mu}$$

and $\Delta_{\mathbf{k}} = \Delta^{(s)}$.

**Pseudospin representation.** The hamiltonian can be written equivalently as

$$h_{\mathbf{k}} = \xi_{\mathbf{k}}\tau^{z} - \mathrm{Re}\{\Delta_{\mathbf{k}}\}\tau^{x} - \mathrm{Im}\{\Delta_{\mathbf{k}}\}\tau^{y}$$

Let now the pseudospin operator $\hat{\mathbf{s}}_{\mathbf{k}}$

$$\hat{s}^{\alpha}_{\mathbf{k}} \equiv \hat{\Phi}^{\dagger}_{\mathbf{k}}\tau^{\alpha}\hat{\Phi}_{\mathbf{k}} \quad \text{with} \quad \hat{\mathbf{s}}_{\mathbf{k}\sigma} = \begin{bmatrix} \hat{s}^{x}_{\mathbf{k}\sigma} \\ \hat{s}^{y}_{\mathbf{k}\sigma} \\ \hat{s}^{z}_{\mathbf{k}\sigma} \end{bmatrix}$$

and the pseudofield $\mathbf{b}_{\mathbf{k}}$:

$$\mathbf{b}_{\mathbf{k}} \equiv \begin{bmatrix} -\mathrm{Re}\{\Delta_{\mathbf{k}}\} \\ -\mathrm{Im}\{\Delta_{\mathbf{k}}\} \\ \xi_{\mathbf{k}} \end{bmatrix}$$

| EV | Meaning |
|---|---|
| $\langle \hat{s}^x_{\mathbf{k}} \rangle$ | $\left\langle \hat{\phi}_{\mathbf{k}} + \hat{\phi}^\dagger_{\mathbf{k}} \right\rangle$ |
| $\langle \hat{s}^y_{\mathbf{k}} \rangle$ | $i \left\langle \hat{\phi}_{\mathbf{k}} - \hat{\phi}^\dagger_{\mathbf{k}} \right\rangle$ |
| $\langle \hat{s}^z_{\mathbf{k}} \rangle$ | $\langle \hat{n}_{\mathbf{k}\uparrow} + \hat{n}_{-\mathbf{k}\downarrow} \rangle - 1$ |

Table 1.3 | List of the various expectation values (EVs) of each pseudospin component and their meaning in terms of fermionic operators.

For a purely real gap, the pseudofield has no $y$ component. In general, different gauges rotate the pseudofield around the $z$ axis.

As was for the AF phase, these operators satisfy a $\mathfrak{su}(2)$ algebra up to an constant, irrelevant in the context of this discussion. They satisfy the identities:

$$\hat{s}^x_{\mathbf{k}} = \hat{\phi}_{\mathbf{k}} + \hat{\phi}^\dagger_{\mathbf{k}} \tag{1.15}$$

$$\hat{s}^y_{\mathbf{k}} = i\left(\hat{\phi}_{\mathbf{k}} - \hat{\phi}^\dagger_{\mathbf{k}}\right) \tag{1.16}$$

$$\hat{s}^z_{\mathbf{k}} = \hat{n}_{\mathbf{k}\uparrow} + \hat{n}_{\mathbf{k}\downarrow} - 1 \tag{1.17}$$

We report the EVs related to these identities Tab. 1.3.

We express the approximated hamiltonian in pseudospin representation as:

$$\hat{H}_t + \hat{H}_U - \mu\hat{N} \simeq \sum_{\mathbf{k}\in\mathrm{BZ}} \mathbf{b}_{\mathbf{k}} \cdot \hat{\mathbf{s}}_{\mathbf{k}} + nU\hat{N} + E^{(\mathrm{SC})}_{\mathrm{a}} - \left[E^{(\mathrm{SC})}_{\mathrm{H}/U} + E^{(\mathrm{SC})}_{\mathrm{C}/U}\right] \tag{1.18}$$

This operator is similar enough to the one found for the AF phase, in Eq. (**??**). It describes a set of pseudospin subject to a uniform field. As for the AF phase, we define the Bogoliubov bands $E_{\mathbf{k}}$

$$E_{\mathbf{k}} \equiv \sqrt{\xi^2_{\mathbf{k}} + |\Delta_{\mathbf{k}}|^2}$$

The key difference with the AF Bogoliubov bands of Eq. (**??**) is that $E_{\mathbf{k}}$ now are functions of $\tilde{\mu}$, since $\xi_{\mathbf{k}}$ depends on the chemical potential. In other words, now the Bogoliubov bands depend on the shifted bands $\xi_{\mathbf{k}}$ instead of the bare bands $\epsilon_{\mathbf{k}}$.

Moreover, let us define the angles $\theta_{\mathbf{k}}$, $\zeta_{\mathbf{k}}$ as

$$\cos 2\theta_{\mathbf{k}} = \frac{\xi_{\mathbf{k}}}{E_{\mathbf{k}}} \qquad \sin 2\theta_{\mathbf{k}} \cos 2\zeta_{\mathbf{k}} = \frac{\mathrm{Re}\{\Delta_{\mathbf{k}}\}}{E_{\mathbf{k}}} \qquad \sin 2\theta_{\mathbf{k}} \sin 2\zeta_{\mathbf{k}} = \frac{\mathrm{Im}\{\Delta_{\mathbf{k}}\}}{E_{\mathbf{k}}}$$

Note that choosing the gauge $\eta = \arg \Delta^{(s)}/2$ the imaginary part of the gap is zero, and this reflects in $2\zeta_{\mathbf{k}} = k \times 2\pi$ with $k \in \mathbb{Z}$.

**Diagonalisation.** For the SC phase, diagonalisation of the HF hamiltonian follows the same procedure as was for the AF phase. Indeed, the diagonalising matrix is given by

$$\begin{aligned} W_{\mathbf{k}} &\equiv e^{-i\theta_{\mathbf{k}}\tau^y} e^{-i\zeta_{\mathbf{k}}\tau^z} \\ &= \begin{bmatrix} \cos\theta_{\mathbf{k}} & -\sin\theta_{\mathbf{k}} \\ \sin\theta_{\mathbf{k}} & \cos\theta_{\mathbf{k}} \end{bmatrix} \begin{bmatrix} e^{-i\zeta_{\mathbf{k}}} & \\ & e^{i\zeta_{\mathbf{k}}} \end{bmatrix} \\ &= \begin{bmatrix} \cos\theta_{\mathbf{k}} e^{-i\zeta_{\mathbf{k}}} & -\sin\theta_{\mathbf{k}} e^{i\zeta_{\mathbf{k}}} \\ \sin\theta_{\mathbf{k}} e^{-i\zeta_{\mathbf{k}}} & \cos\theta_{\mathbf{k}} e^{i\zeta_{\mathbf{k}}} \end{bmatrix} \end{aligned} \tag{1.19}$$

and the graphic procedure is the same of Fig. **??**, with $\epsilon \to \xi$. We get:

$$d_{\mathbf{k}} = W_{\mathbf{k}} h_{\mathbf{k}} W^\dagger_{\mathbf{k}} \quad \text{with} \quad d_{\mathbf{k}} \equiv \begin{bmatrix} E_{\mathbf{k}} & \\ & -E_{\mathbf{k}} \end{bmatrix} \tag{1.20}$$

In the tilted frame we have the expression for the Nambu spinors:

$$\hat{\Gamma}_{\mathbf{k}} = W_{\mathbf{k}} \hat{\Psi}_{\mathbf{k}} \tag{1.21}$$

Explicitly:

$$\begin{aligned}\hat{\Gamma}_{\mathbf{k}} &= \begin{bmatrix}\cos\theta_{\mathbf{k}}e^{-i\zeta_{\mathbf{k}}} & -\sin\theta_{\mathbf{k}}e^{i\zeta_{\mathbf{k}}}\\ \sin\theta_{\mathbf{k}}e^{-i\zeta_{\mathbf{k}}} & \cos\theta_{\mathbf{k}}e^{i\zeta_{\mathbf{k}}}\end{bmatrix}\begin{bmatrix}\hat{c}_{\mathbf{k}\uparrow}\\ \hat{c}^{\dagger}_{-\mathbf{k}\downarrow}\end{bmatrix}\\ &= \begin{bmatrix}\cos\theta_{\mathbf{k}}e^{-i\zeta_{\mathbf{k}}}\hat{c}_{\mathbf{k}\uparrow} - \sin\theta_{\mathbf{k}}e^{i\zeta_{\mathbf{k}}}\hat{c}^{\dagger}_{-\mathbf{k}\downarrow}\\ \sin\theta_{\mathbf{k}}e^{-i\zeta_{\mathbf{k}}}\hat{c}_{\mathbf{k}\uparrow} + \cos\theta_{\mathbf{k}}e^{i\zeta_{\mathbf{k}}}\hat{c}^{\dagger}_{-\mathbf{k}\downarrow}\end{bmatrix} \equiv \begin{bmatrix}\hat{\gamma}^{(+)}_{\mathbf{k}}\\ \hat{\gamma}^{(-)}_{\mathbf{k}}\end{bmatrix}\end{aligned} \tag{1.22}$$

The new Fermionic operators describe a Fermi gas on distributed on two mirrored bands,

$$\hat{H}_t + \hat{H}_U - \tilde{\mu}\hat{N} \simeq \sum_{\mathbf{k}\in\mathrm{BZ}}\sum_{\sigma} E_{\mathbf{k}}\left(\hat{\gamma}^{(+)\dagger}_{\mathbf{k}}\hat{\gamma}^{(+)}_{\mathbf{k}} - \hat{\gamma}^{(-)\dagger}_{\mathbf{k}}\hat{\gamma}^{(-)}_{\mathbf{k}}\right) + (\cdots)$$

with $(\cdots)$ the various energy shifts.

### 1.2.6 Features of the SC solution

In this section we give some details about the HF solution for the translationally-invariant SC phase. We prove that in the conventional HM, with local repulsion $\hat{H}_U$, isotropic BCS-like superconductivity is forbidden. This serves as ground to motivate the passage to the EHM.

**Pseudospin EVs.** The EVs of pseudospin operators in the various directions are given by:

$$\begin{aligned}\langle\hat{\Psi}^{\dagger}_{\mathbf{k}}\tau^x\hat{\Psi}_{\mathbf{k}}\rangle &= -\sin 2\theta_{\mathbf{k}}\cos 2\zeta_{\mathbf{k}}\langle\hat{\Gamma}^{\dagger}_{\mathbf{k}}\tau^z\hat{\Gamma}_{\mathbf{k}}\rangle\\ \langle\hat{\Psi}^{\dagger}_{\mathbf{k}}\tau^y\hat{\Psi}_{\mathbf{k}}\rangle &= -\sin 2\theta_{\mathbf{k}}\sin 2\zeta_{\mathbf{k}}\langle\hat{\Gamma}^{\dagger}_{\mathbf{k}}\tau^z\hat{\Gamma}_{\mathbf{k}}\rangle\\ \langle\hat{\Psi}^{\dagger}_{\mathbf{k}}\tau^z\hat{\Psi}_{\mathbf{k}}\rangle &= \cos 2\theta_{\mathbf{k}}\langle\hat{\Gamma}^{\dagger}_{\mathbf{k}}\tau^z\hat{\Gamma}_{\mathbf{k}}\rangle\end{aligned}$$

In the tilted frame, the averaged $z$ projection of pseudospin is:

$$\langle\hat{\Gamma}^{\dagger}_{\mathbf{k}}\tau^z\hat{\Gamma}_{\mathbf{k}}\rangle = f\left(E_{\mathbf{k}};\beta,0\right) - f\left(-E_{\mathbf{k}};\beta,0\right) \tag{1.23}$$

Compared with the respective relation for the AF phase, Eq. (**??**), the main difference here is that the chemical potential inside the Fermi-Dirac distribution is set to zero. This is due to the fact that $\tilde{\mu}$ is already included in the renormalised bands $E_{\mathbf{k}}$. The approximated quadratic hamiltonian describes a gas of $\gamma$ fermions at null chemical potential.

Let us define the thermal factors:

$$\mathrm{Th}^{(\pm)}_{\mathbf{k}}[\beta] \equiv f\left(-E_{\mathbf{k}};\beta,0\right) \pm f\left(E_{\mathbf{k}};\beta,0\right) \tag{1.24}$$

In the end we have the EVs:

$$\langle\hat{s}^x_{\mathbf{k}}\rangle = \frac{\mathrm{Re}\{\Delta_{\mathbf{k}}\}}{E_{\mathbf{k}}}\mathrm{Th}^{(-)}_{\mathbf{k}}[\beta] \tag{1.25}$$

$$\langle\hat{s}^y_{\mathbf{k}}\rangle = \frac{\mathrm{Im}\{\Delta_{\mathbf{k}}\}}{E_{\mathbf{k}}}\mathrm{Th}^{(-)}_{\mathbf{k}}[\beta] \tag{1.26}$$

$$\langle\hat{s}^z_{\mathbf{k}}\rangle = -\frac{\xi_{\mathbf{k}}}{E_{\mathbf{k}}}\mathrm{Th}^{(-)}_{\mathbf{k}}[\beta] \tag{1.27}$$

Note that, as is expressed in Tab. 1.3, the $z$ component of the pseudospin is related to density, being $1 + \langle\hat{s}^z_{\mathbf{k}}\rangle = \langle\hat{n}_{\mathbf{k}\uparrow} + \hat{n}_{-\mathbf{k}\downarrow}\rangle$.

***s*-wave gap self-consistency equation.** Applying the HF method to the conventional HM in SC phase, we obtained just one HFP: $w^{(s)}$. Let us derive its self-consistency equation. First, let

$$\tau^{\pm} \equiv \frac{\tau^x \pm i\tau^y}{2}$$

Recall Eq. (1.8) and expand it, to obtain

$$
\begin{aligned}
w^{(s)} &= \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \langle \hat{\phi}^\dagger_{\mathbf{k}} \rangle \\
&= \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \langle \hat{\Phi}^\dagger_{\mathbf{k}} \tau^+ \hat{\Phi}_{\mathbf{k}} \rangle \\
&= \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \frac{\langle \hat{s}^x_{\mathbf{k}} \rangle + i \langle \hat{s}^y_{\mathbf{k}} \rangle}{2} \\
&= \frac{1}{2L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \frac{\Delta_{\mathbf{k}}}{E_{\mathbf{k}}} \mathrm{Th}^{(-)}_{\mathbf{k}}[\beta]
\end{aligned} \tag{1.28}
$$

Last line is the explicit form of the self-consistency equation for $w^{(s)}$.

**Impossibility of $s$-wave superconductivity.** Let us define the critical $U$,

$$
U_\mathrm{c}[\beta] \equiv \left[ \frac{1}{2L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \frac{\mathrm{Th}^{(-)}_{\mathbf{k}}[\beta]}{E_{\mathbf{k}}} \right]^{-1} \tag{1.29}
$$

We call it "critical" because its value gives a sort of boundary for convergence. Indeed, as we show in App. ??, for the SC phase in conventional HM, we are able to optimise parametrically the mixing parameter $g$ in order to ensure convergence, based on the ratio $U/U_\mathrm{c}[\beta]$.

Let us show that superconductivity is not possible in the conventional HM with $U > 0$. Eq. (1.28) can be rewritten as:

$$
w^{(s)} = -\frac{U}{U_\mathrm{c}[\beta]} w^{(s)}
$$

By construction $U_\mathrm{c}[\beta] > 0$ (the thermal factor is negative-defined). Hence the unique possible self-consistent solution for the equation above is $w^{(s)} = 0$. This implication terminates the argument: superconductivity cannot establish self-consistently at HF level in the repulsive HM. If instead $U < 0$, which is, if the model is attractive, we the aforementioned implication is not true and a self-consistent non-zero solution is possible.

**Density.** In order to estimate density, we can use the $z$ projection of the pseudospin operators. From Eq. (1.17) we know

$$
\hat{s}^z_{\mathbf{k}} + 1 = \hat{n}_{\mathbf{k}\uparrow} + \hat{n}_{\mathbf{k}\downarrow}
$$

Then, density can be estimated as:

$$
\begin{aligned}
n &= \frac{1}{2L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \langle \hat{n}_{\mathbf{k}\uparrow} + \hat{n}_{-\mathbf{k}\downarrow} \rangle \\
&= \frac{1}{2L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \left( \langle \hat{s}^z_{\mathbf{k}} \rangle + 1 \right) \\
&= \frac{1}{2L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \left( 1 - \frac{\xi_{\mathbf{k}}}{E_{\mathbf{k}}} \mathrm{Th}^{(-)}_{\mathbf{k}}[\beta] \right)
\end{aligned} \tag{1.30}
$$

We can now move to the EHM.

## 1.3 Extension to the EHM

Add now the non-local interaction $\hat{H}_V$. As we did for the AF phase, we look for a solution with an identical mathematical structure as the one here described for the conventional HM. We define the renormalised quantities

$$
\xi_{\mathbf{k}} \to \tilde{\xi}_{\mathbf{k}} \qquad E_{\mathbf{k}} \to \tilde{E}_{\mathbf{k}} \qquad \Delta_{\mathbf{k}} \to \tilde{\Delta}_{\mathbf{k}} \qquad \mu \to \tilde{\mu}
$$

To be completely correct, the chemical potential was already renormalized in the HM derivation in Sec. **??**. It here acquires yet another renormalization. The band energies renormalization is simply

$$\tilde{E}_{\mathbf{k}} \equiv \sqrt{\tilde{\xi}_{\mathbf{k}}^2 + |\tilde{\Delta}_{\mathbf{k}}|^2}$$

Let the renormalised angle:

$$\sin 2\tilde{\theta}_{\mathbf{k}} = \frac{|\tilde{\Delta}_{\mathbf{k}}|}{\tilde{E}_{\mathbf{k}}} \qquad \cos 2\tilde{\theta}_{\mathbf{k}} = \frac{\tilde{\xi}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}} \tag{1.31}$$

and similarly the angle $\tilde{\zeta}_{\mathbf{k}}$,

$$\sin 2\tilde{\zeta}_{\mathbf{k}} = \frac{\mathrm{Re}\{\tilde{\Delta}_{\mathbf{k}}\}}{|\tilde{\Delta}_{\mathbf{k}}|} \qquad \cos 2\tilde{\zeta}_{\mathbf{k}} = \frac{\mathrm{Im}\{\tilde{\Delta}_{\mathbf{k}}\}}{|\tilde{\Delta}_{\mathbf{k}}|} \tag{1.32}$$

The pseudospin EVs are renormalised too,

$$\langle \hat{\Phi}^{\dagger}_{\mathbf{k}} \tau^x \hat{\Phi}_{\mathbf{k}} \rangle = -\sin 2\tilde{\theta}_{\mathbf{k}} \sin 2\tilde{\zeta}_{\mathbf{k}} \langle \hat{\Gamma}^{\dagger}_{\mathbf{k}} \tau^z \hat{\Gamma}_{\mathbf{k}} \rangle \tag{1.33}$$

$$\langle \hat{\Phi}^{\dagger}_{\mathbf{k}} \tau^y \hat{\Phi}_{\mathbf{k}} \rangle = -\sin 2\tilde{\theta}_{\mathbf{k}} \cos 2\tilde{\zeta}_{\mathbf{k}} \langle \hat{\Gamma}^{\dagger}_{\mathbf{k}} \tau^z \hat{\Gamma}_{\mathbf{k}} \rangle \tag{1.34}$$

$$\langle \hat{\Phi}^{\dagger}_{\mathbf{k}} \tau^z \hat{\Phi}_{\mathbf{k}} \rangle = \cos 2\tilde{\theta}_{\mathbf{k}} \langle \hat{\Gamma}^{\dagger}_{\mathbf{k}} \tau^z \hat{\Gamma}_{\mathbf{k}} \rangle \tag{1.35}$$

Let also:

$$\langle \hat{\Gamma}^{\dagger}_{\mathbf{k}} \tau^z \hat{\Gamma}_{\mathbf{k}} \rangle = f\left(\tilde{E}_{\mathbf{k}}; \beta, 0\right) - f\left(-\tilde{E}_{\mathbf{k}}; \beta, 0\right) \tag{1.36}$$

$$\tilde{\mathrm{Th}}^{(\pm)}_{\mathbf{k}}[\beta, \tilde{\mu}] \equiv f\left(-\tilde{E}_{\mathbf{k}}; \beta, 0\right) \pm f\left(\tilde{E}_{\mathbf{k}}; \beta, 0\right) \tag{1.37}$$

For the SC phase, the thermal factors reduce to:

$$\tilde{\mathrm{Th}}^{(+)}_{\mathbf{k}} = 1 \tag{1.38}$$

$$\tilde{\mathrm{Th}}^{(-)}_{\mathbf{k}} = -\tanh\left(\frac{\beta \tilde{E}_{\mathbf{k}}}{2}\right) \tag{1.39}$$

Hence, pseudospin expectation values are:

$$\langle \hat{s}^x_{\mathbf{k}} \rangle = -\frac{\mathrm{Re}\{\tilde{\Delta}_{\mathbf{k}}\}}{\tilde{E}_{\mathbf{k}}} \tanh\left(\frac{\beta \tilde{E}_{\mathbf{k}}}{2}\right) \tag{1.40}$$

$$\langle \hat{s}^y_{\mathbf{k}} \rangle = -\frac{\mathrm{Im}\{\tilde{\Delta}_{\mathbf{k}}\}}{\tilde{E}_{\mathbf{k}}} \tanh\left(\frac{\beta \tilde{E}_{\mathbf{k}}}{2}\right) \tag{1.41}$$

$$\langle \hat{s}^z_{\mathbf{k}} \rangle = \frac{\tilde{\xi}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}} \tanh\left(\frac{\beta \tilde{E}_{\mathbf{k}}}{2}\right) \tag{1.42}$$

These equations are an extension of Eqns. (1.40), (1.41), (1.42) for the EHM.

We can rewrite the equations for $w^{(s)}$ and $n$ accordingly,

$$w^{(s)} = -\frac{1}{L_x L_y} \sum_{\mathbf{k} \in \mathrm{BZ}} \frac{\tilde{\Delta}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}} \tanh\left(\frac{\beta \tilde{E}_{\mathbf{k}}}{2}\right) \tag{1.43}$$

$$n = \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \mathrm{BZ}} \left[1 + \cos 2\tilde{\theta}_{\mathbf{k}} \tanh\left(\frac{\beta \tilde{E}_{\mathbf{k}}}{2}\right)\right] \tag{1.44}$$

where we used Eqns. (1.28), (1.30).

From Tabs. **??** and **??** we know none of the three channels (Hartree, Fock, Cooper) is subject to selection rules. Thus all terms of the decomposition of $\hat{H}_V$ given in Eq. (**??**) contribute. As always, we break down each.

## 1.4 Non-local Hartree effects: chemical potential renormalisation

We start by analysing the non-local Hartree channel,

$$\hat{H}_V \simeq -V \sum_{\langle ij \rangle} \sum_{\sigma,\sigma'} \Big( \hat{c}^\dagger_{i\sigma} \hat{c}_{i\sigma} \big\langle \hat{c}^\dagger_{j\sigma'} \hat{c}_{j\sigma'} \big\rangle + \big\langle \hat{c}^\dagger_{i\sigma} \hat{c}_{i\sigma} \big\rangle \hat{c}^\dagger_{j\sigma'} \hat{c}_{j\sigma'} - \big\langle \hat{c}^\dagger_{i\sigma} \hat{c}_{i\sigma} \big\rangle \big\langle \hat{c}^\dagger_{j\sigma'} \hat{c}_{j\sigma'} \big\rangle \Big) \\ + (\text{Fock channel}) + (\text{Cooper channel})$$

where we used Eq. (**??**). This discussion is based on that for the normal phase in Sec. **??**.

### 1.4.1 Non-local Hartree channel in the SC phase: analytic derivation

Without repeating the calculations, the non-local attraction $\hat{H}_V$ is reduced in the Hartree channel to

$$\hat{H}_V \simeq -2nzV \times \hat{N} - E^{(\mathrm{SC})}_{\mathrm{H}/V} + (\text{all the rest})$$

This, together with the $\hat{H}_U$ Hartree channel decomposition

$$\hat{H}_U \simeq nU \times \hat{N} - E^{(\mathrm{SC})}_{\mathrm{H}/U}$$

gives the chemical potential full renormalization, identical to Eq. (**??**),

$$\tilde{\mu} \equiv \mu + n(2zV - U) \tag{1.45}$$

### 1.4.2 Non-local Hartree contraction energy

The contraction energy for the non-local sector was derived in Eq. (**??**),

$$E^{(\mathrm{SC})}_{\mathrm{H}/V} = -L_x L_y \times 2zn^2 V$$

and together with the local counterpart of Eq. (**??**)

$$E^{(\mathrm{SC})}_{\mathrm{H}/U} = L_x L_y \times n^2 U$$

Now we move to the more interesting Fock sector.

## 1.5 Fock effects: bare bands renormalization

As done for the other phases, we here discuss the bands renormalization effect derived from the Fock sector of the non-local attraction. From Eq. (**??**), we have

$$\hat{H}_V \simeq (\text{Hartree channel}) \\ V \sum_{\langle ij \rangle} \sum_{\sigma,\sigma'} \Big( \hat{c}^\dagger_{i\sigma} \hat{c}_{j\sigma'} \big\langle \hat{c}^\dagger_{j\sigma'} \hat{c}_{i\sigma} \big\rangle + \big\langle \hat{c}^\dagger_{i\sigma} \hat{c}_{j\sigma'} \big\rangle \hat{c}^\dagger_{j\sigma'} \hat{c}_{i\sigma} - \big\langle \hat{c}^\dagger_{i\sigma} \hat{c}_{j\sigma'} \big\rangle \big\langle \hat{c}^\dagger_{j\sigma'} \hat{c}_{i\sigma} \big\rangle \Big) + (\text{Cooper channel})$$

This discussion is essentially analogous the one done for the normal phase in Sec. **??**. As for previous phases, following the selection rules of Tab. **??**, only the s.s. channel contributes. Therefore we use from now on, in this section, $\sigma = \sigma'$.

### 1.5.1 Fock channel in the SC phase: analytic derivation

Let us define, as done for the normal phase in Eq. (**??**) and for the AF phase in Eq. (**??**),

$$u_{\mathbf{k}\sigma} \equiv \langle \hat{c}^\dagger_{\mathbf{k}\sigma} \hat{c}_{\mathbf{k}\sigma} \rangle \tag{1.46}$$

Eq. (**??**) stands perfectly valid, translated to the SC phase

$$V\sum_{\langle ij\rangle}\sum_{\sigma}\left[\langle\hat{c}^{\dagger}_{i\sigma}\hat{c}_{j\sigma}\rangle\hat{c}^{\dagger}_{j\sigma}\hat{c}_{i\sigma}+\hat{c}^{\dagger}_{i\sigma}\hat{c}_{j\sigma}\langle\hat{c}^{\dagger}_{j\sigma}\hat{c}_{i\sigma}\rangle-\langle\hat{c}^{\dagger}_{i\sigma}\hat{c}_{j\sigma}\rangle\langle\hat{c}^{\dagger}_{j\sigma}\hat{c}_{i\sigma}\rangle\right]$$
$$\overset{\mathcal{F}}{=}\sum_{\gamma,\sigma}\left[\frac{1}{L_xL_y}\sum_{\mathbf{k}\in\mathrm{BZ}}\varphi^{(\gamma)}_{\mathbf{k}}u_{\mathbf{k}\sigma}\right]\times V\sum_{\mathbf{k}'\in\mathrm{BZ}}\varphi^{(\gamma)*}_{\mathbf{k}'}\hat{c}^{\dagger}_{\mathbf{k}'\sigma}\hat{c}_{\mathbf{k}'\sigma}-E^{(\mathrm{SC})}_{\mathrm{F}/V} \tag{1.47}$$

being

$$E^{(\mathrm{SC})}_{\mathrm{F}/V}\equiv\frac{V}{L_xL_y}\sum_{\mathbf{k},\mathbf{k}'}\sum_{\sigma}\left[\cos\left(\delta k_x\right)+\cos\left(\delta k_y\right)\right]\langle\hat{c}^{\dagger}_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}\sigma}\rangle\langle\hat{c}^{\dagger}_{\mathbf{k}'\sigma}\hat{c}_{\mathbf{k}'\sigma}\rangle \tag{1.48}$$

To get to this point, we employed the assumption

$$\langle\hat{c}^{\dagger}_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}'\sigma'}\rangle=\delta_{\mathbf{k}=\mathbf{k}'}\delta_{\sigma=\sigma'}\langle\hat{c}^{\dagger}_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}\sigma}\rangle$$

which essentially states translational invariance and spin degeneracy. This assumption is verified by the BCS ground state of Eq. **??**. Let us define each $\gamma$-wave component as

$$u^{(\gamma)}_{\sigma}=\frac{1}{L_xL_y}\sum_{\mathbf{k}\in\mathrm{BZ}}\varphi^{(\gamma)}_{\mathbf{k}}\langle\hat{c}^{\dagger}_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}\sigma}\rangle$$

These are our HFPs in the Fock channel. The SC phase does not break inversion symmetry: then, as we concluded for the other phases, $p_\ell$ components are null. The renormalised bands are, as in Eqns. (**??**), (**??**),

$$\tilde{\epsilon}_{\mathbf{k}}\equiv\epsilon_{\mathbf{k}}+u^{(s^*)}V(\cos k_x+\cos k_y)+u^{(d)}V(\cos k_x-\cos k_y)$$

Let us compute the EVs of $u_{\mathbf{k}\sigma}$ in the two spin sectors,

$$\begin{aligned}u_{\mathbf{k}\uparrow}&=\langle\hat{c}^{\dagger}_{\mathbf{k}\uparrow}\hat{c}_{\mathbf{k}\uparrow}\rangle\\&=\left\langle\hat{\Phi}^{\dagger}_{\mathbf{k}}\frac{\mathbb{I}+\tau^z}{2}\hat{\Phi}_{\mathbf{k}}\right\rangle\\&=\frac{1}{2}\left[\left\langle\hat{\Gamma}^{\dagger}_{\mathbf{k}}\hat{\Gamma}_{\mathbf{k}}\right\rangle+\langle\hat{s}^z_{\mathbf{k}}\rangle\right]\\&=\frac{1}{2}\left[\tilde{\mathrm{Th}}^{(+)}_{\mathbf{k}}[\beta]-\cos(2\tilde{\theta}_{\mathbf{k}})\tilde{\mathrm{Th}}^{(-)}_{\mathbf{k}}[\beta]\right]\\&=\frac{1}{2}\left[1-\cos(2\tilde{\theta}_{\mathbf{k}})\tanh\left(\frac{\beta\tilde{E}_{\mathbf{k}}}{2}\right)\right]\end{aligned} \tag{1.49}$$

having we used Eq. (**??**) and following. For the spin $\downarrow$ EV,

$$\begin{aligned}u_{-\mathbf{k}\downarrow}&=\langle\hat{c}^{\dagger}_{-\mathbf{k}\downarrow}\hat{c}_{-\mathbf{k}\downarrow}\rangle\\&=1-\langle\hat{c}_{-\mathbf{k}\downarrow}\hat{c}^{\dagger}_{-\mathbf{k}\downarrow}\rangle\\&=1-\left\langle\hat{\Phi}^{\dagger}_{\mathbf{k}}\frac{\mathbb{I}-\tau^z}{2}\hat{\Phi}_{\mathbf{k}}\right\rangle\\&=1-\frac{1}{2}\left[\left\langle\hat{\Gamma}^{\dagger}_{\mathbf{k}}\hat{\Gamma}_{\mathbf{k}}\right\rangle-\langle\hat{s}^z_{\mathbf{k}}\rangle\right]\\&=1-\frac{1}{2}\left[\tilde{\mathrm{Th}}^{(+)}_{\mathbf{k}}[\beta]+\cos 2\tilde{\theta}_{\mathbf{k}}\tilde{\mathrm{Th}}^{(-)}_{\mathbf{k}}[\beta]\right]\\&=1-\frac{1}{2}\left[1+\cos(2\tilde{\theta}_{\mathbf{k}})\tanh\left(\frac{\beta\tilde{E}_{\mathbf{k}}}{2}\right)\right]\end{aligned} \tag{1.50}$$

From the above results, note first that

$$u_{\mathbf{k}\uparrow}=u_{-\mathbf{k}\downarrow}$$

| **Symmetry** | **Self-consistency equation** | **Graph** |
|---|---|---|
| $s^*$-wave | $u^{(s^*)} = \dfrac{1}{2L_xL_y}\sum_{\mathbf{k}\in\mathrm{BZ}}(\cos k_x + \cos k_y)\left[1 - \dfrac{\tilde{\xi}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}}\tanh\left(\dfrac{\beta\tilde{E}_{\mathbf{k}}}{2}\right)\right]$ | Fig. **??** |
| $d_{x^2-y^2}$-wave | $u^{(d)} = \dfrac{1}{2L_xL_y}\sum_{\mathbf{k}\in\mathrm{BZ}}(\cos k_x - \cos k_y)\left[1 - \dfrac{\tilde{\xi}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}}\tanh\left(\dfrac{\beta\tilde{E}_{\mathbf{k}}}{2}\right)\right]$ | Fig. **??** |

Table 1.4 | Symmetry resolved self-consistency equations for the harmonics of $u_{\mathbf{k}}$.

which is coherent with our requirement of a uniform-density, spin uniform and inversion symmetric phase. Note also that Eqns. (1.49), (1.50) combine to give the expression for density of Eq. (1.44).

If we take the $\tilde{\theta}_{\mathbf{k}} \to 0$ limit, we must retrieve the Fermi-Dirac distribution. Indeed, this limit coincides with $U \to 0$. Explicitly:

$$\begin{aligned}
\lim_{\tilde{\theta}_{\mathbf{k}}\to 0} u_{\mathbf{k}\sigma} &= \frac{1}{2}\left[1 + \tanh\left(\frac{\beta\tilde{\xi}_{\mathbf{k}}}{2}\right)\right] \\
&= \frac{1}{2}\left[\frac{e^{\beta\tilde{\xi}_{\mathbf{k}}/2} + e^{-\beta\tilde{\xi}_{\mathbf{k}}/2}}{e^{\beta\tilde{\xi}_{\mathbf{k}}/2} + e^{-\beta\tilde{\xi}_{\mathbf{k}}/2}} - \frac{e^{\beta\tilde{\xi}_{\mathbf{k}}/2} - e^{-\beta\tilde{\xi}_{\mathbf{k}}/2}}{e^{\beta\tilde{\xi}_{\mathbf{k}}/2} + e^{-\beta\tilde{\xi}_{\mathbf{k}}/2}}\right] \\
&= f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu})
\end{aligned}$$

Correctly, the null-gap limit (for which the angle reduces to zero) gives back a perfectly degenerate Fermi gas.

We can then derive the $u_{\mathbf{k}}$-related HFPs self-consistency equations. In next passages, we use $\sigma = \uparrow$ but could have used the opposite due to spin degeneracy discussed above. Then:

$$\begin{aligned}
u^{(\gamma)} &\equiv \frac{1}{L_xL_y}\sum_{\mathbf{k}\in\mathrm{BZ}} \varphi_{\mathbf{k}}^{(\gamma)}\langle \hat{n}_{\mathbf{k}\uparrow}\rangle \\
&= \frac{1}{2L_xL_y}\sum_{\mathbf{k}\in\mathrm{BZ}} \varphi_{\mathbf{k}}^{(\gamma)}\left[1 - \cos(2\tilde{\theta}_{\mathbf{k}})\tanh\left(\frac{\beta\tilde{E}_{\mathbf{k}}}{2}\right)\right]
\end{aligned} \tag{1.51}$$

reported explicitly for the only two inversion symmetric structures in Tab. 1.4.

### 1.5.2 Fock contraction energy

In terms of the energy shift due to Wick's contraction, Eq. (**??**) holds identically

$$E_{\mathrm{F}/V}^{(\mathrm{SC})} = L_xL_y \times V\left[\left|u^{(s^*)}\right|^2 + \left|u^{(d)}\right|^2\right]$$

with the only obvious difference that the $u^{(\gamma)}$ are computed on the SC ground state by the means of Eqns. (1.49), (1.50).

## 1.6 Non-local Cooper effects: anisotropic pairing

In this section we deal with the Cooper part of Eq. (**??**),

$$\begin{aligned}
\hat{H}_V =\simeq &\ (\text{Hartree channel}) + (\text{Fock channel}) \\
&- V\sum_{\langle ij\rangle}\sum_{\sigma,\sigma'}\left(\hat{c}^\dagger_{i\sigma}\hat{c}^\dagger_{j\sigma'}\langle\hat{c}_{j\sigma'}\hat{c}_{i\sigma}\rangle + \langle\hat{c}^\dagger_{i\sigma}\hat{c}^\dagger_{j\sigma'}\rangle\hat{c}_{j\sigma'}\hat{c}_{i\sigma} - \langle\hat{c}^\dagger_{i\sigma}\hat{c}^\dagger_{j\sigma'}\rangle\langle\hat{c}_{j\sigma'}\hat{c}_{i\sigma}\rangle\right)
\end{aligned}$$

We know from Eq. (**??**) that the non-local attraction can be separated in o.s. and s.s. channels. As we reported in Tab. **??** and stated in Sec. 1.1, the only non-zero contribution comes from the o.s. sector.

### 1.6.1 Non-local Cooper o.s. channel in the SC phase: analytic derivation

The o.s. Cooper sector contributes both to singlet and triplet pairing. Note that in singlet pairing channel the spatial wavefunction must be inversion-symmetric. On the contrary, in triplet pairing channel it needs to be inversion-antisymmetric.

We move to reciprocal space. From Eq. (**??**), we get

$$\hat{H}_V^{(\mathrm{o.s.})} \simeq (\text{Hartree channel}) \\ - \frac{2V}{L_x L_y} \sum_{\mathbf{K},\mathbf{k},\mathbf{k}'} \left[\cos(\delta k_x) + \cos(\delta k_y)\right] \times \Big[\langle \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\uparrow} \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\downarrow} \rangle \hat{c}_{\mathbf{K}-\mathbf{k}'\downarrow} \hat{c}_{\mathbf{K}+\mathbf{k}'\uparrow} \\ + \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\uparrow} \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\downarrow} \langle \hat{c}_{\mathbf{K}-\mathbf{k}'\downarrow} \hat{c}_{\mathbf{K}+\mathbf{k}'\uparrow} \rangle - \langle \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\uparrow} \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\downarrow} \rangle \langle \hat{c}_{\mathbf{K}-\mathbf{k}'\downarrow} \hat{c}_{\mathbf{K}+\mathbf{k}'\uparrow} \rangle \Big]$$

Identical considerations as for the local repulsion hold, and uniquely the $\mathbf{K} = \mathbf{0}$ term gives a non-zero contribution, in general. Therefore:

$$\hat{H}_V^{(\mathrm{o.s.})} \simeq (\text{Hartree channel}) \\ - \frac{2V}{L_x L_y} \sum_{\mathbf{k},\mathbf{k}'} \left[\cos(\delta k_x) + \cos(\delta k_y)\right] \times \Big[\langle \hat{c}^\dagger_{\mathbf{k}\uparrow} \hat{c}^\dagger_{-\mathbf{k}\downarrow} \rangle \hat{c}_{-\mathbf{k}'\downarrow} \hat{c}_{\mathbf{k}'\uparrow} \\ + \hat{c}^\dagger_{\mathbf{k}\uparrow} \hat{c}^\dagger_{-\mathbf{k}\downarrow} \langle \hat{c}_{-\mathbf{k}'\downarrow} \hat{c}_{\mathbf{k}'\uparrow} \rangle - \langle \hat{c}^\dagger_{\mathbf{k}\uparrow} \hat{c}^\dagger_{-\mathbf{k}\downarrow} \rangle \langle \hat{c}_{-\mathbf{k}'\downarrow} \hat{c}_{\mathbf{k}'\uparrow} \rangle \Big]$$

Then, using the harmonics decomposition of Eq. (**??**), we end up with:

$$\hat{H}_V^{(\mathrm{o.s.})} \simeq (\text{Hartree channel}) - \frac{V}{L_x L_y} \sum_\gamma \sum_{\mathbf{k},\mathbf{k}'} \varphi^{(\gamma)}_{\mathbf{k}} \varphi^{(\gamma)*}_{\mathbf{k}'} \left[ w_{\mathbf{k}} \hat{\phi}_{\mathbf{k}'} + w^*_{\mathbf{k}} \hat{\phi}^\dagger_{\mathbf{k}'} \right] - E^{(\mathrm{SC})}_{\mathrm{C}/V/\mathrm{o.s.}} \tag{1.52}$$

In the equation above the contraction energy is given by

$$E^{(\mathrm{SC})}_{\mathrm{C}/V/\mathrm{o.s.}} \equiv -\frac{V}{L_x L_y} \sum_\gamma \sum_{\mathbf{k},\mathbf{k}'} \varphi^{(\gamma)}_{\mathbf{k}} \varphi^{(\gamma)*}_{\mathbf{k}'} w_{\mathbf{k}} w^*_{\mathbf{k}'} \tag{1.53}$$

Then, let the harmonics $w^{(\gamma)}$ be

$$w^{(\gamma)} = \frac{1}{L_x L_y} \sum_{\mathbf{k}\in \mathrm{BZ}} \varphi^{(\gamma)}_{\mathbf{k}} w_{\mathbf{k}}$$

Eq. (1.52) becomes

$$\hat{H}_V^{(\mathrm{o.s.})} \simeq (\text{Hartree channel}) - V \sum_\gamma \sum_{\mathbf{k}\in \mathrm{BZ}} \left[ w^{(\gamma)} \varphi^{(\gamma)*}_{\mathbf{k}} \hat{\phi}_{\mathbf{k}} + \text{h.c.} \right] - E^{(\mathrm{SC})}_{\mathrm{C}/V/\mathrm{o.s.}}$$

extending the symmetry sum to $\gamma \in \{s^*, p_x, p_y, d_{x^2-y^2}\}$.

For the $\gamma$-wave component of the superconducting order parameter, the following chain, similar to Eq. (1.28), holds:

$$\begin{aligned} w^{(\gamma)} &\equiv \frac{1}{L_x L_y} \sum_{\mathbf{k}\in \mathrm{BZ}} \varphi^{(\gamma)}_{\mathbf{k}} \langle \hat{\phi}^\dagger_{\mathbf{k}} \rangle \\ &= \frac{1}{L_x L_y} \sum_{\mathbf{k}\in \mathrm{BZ}} \varphi^{(\gamma)}_{\mathbf{k}} \langle \hat{\Phi}^\dagger_{\mathbf{k}} \tau^+ \hat{\Phi}_{\mathbf{k}} \rangle \\ &= \frac{1}{L_x L_y} \sum_{\mathbf{k}\in \mathrm{BZ}} \varphi^{(\gamma)}_{\mathbf{k}} \frac{\langle \hat{s}^x_{\mathbf{k}} \rangle + i \langle \hat{s}^y_{\mathbf{k}} \rangle}{2} \\ &= \frac{1}{2 L_x L_y} \sum_{\mathbf{k}\in \mathrm{BZ}} \varphi^{(\gamma)}_{\mathbf{k}} \frac{\tilde{\Delta}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}} \mathrm{Th}^{(-)}_{\mathbf{k}}[\beta] \end{aligned} \tag{1.54}$$

| Symmetry | Self-consistency equation | Graph |
|---|---|---|
| $s$-wave | $w^{(s)} = \frac{1}{2L_xL_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \frac{\tilde{\Delta}_\mathbf{k}}{\tilde{E}_\mathbf{k}} \tanh\left(\frac{\beta\tilde{E}_\mathbf{k}}{2}\right)$ | Fig. ?? |
| $s^*$-wave | $w^{(s^*)} = \frac{1}{2L_xL_y} \sum_{\mathbf{k}\in\mathrm{BZ}} (\cos k_x + \cos k_y) \frac{\tilde{\Delta}_\mathbf{k}}{\tilde{E}_\mathbf{k}} \tanh\left(\frac{\beta\tilde{E}_\mathbf{k}}{2}\right)$ | Fig. ?? |
| $p_x$-wave | $w^{(p_x)} = \frac{i\sqrt{2}}{2L_xL_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \sin k_x \frac{\tilde{\Delta}_\mathbf{k}}{\tilde{E}_\mathbf{k}} \tanh\left(\frac{\beta\tilde{E}_\mathbf{k}}{2}\right)$ | Fig. ?? |
| $p_y$-wave | $w^{(p_y)} = \frac{i\sqrt{2}}{2L_xL_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \sin k_y \frac{\tilde{\Delta}_\mathbf{k}}{\tilde{E}_\mathbf{k}} \tanh\left(\frac{\beta\tilde{E}_\mathbf{k}}{2}\right)$ | Fig. ?? |
| $d_{x^2-y^2}$-wave | $w^{(d)} = \frac{1}{2L_xL_y} \sum_{\mathbf{k}\in\mathrm{BZ}} (\cos k_x - \cos k_y) \frac{\tilde{\Delta}_\mathbf{k}}{\tilde{E}_\mathbf{k}} \tanh\left(\frac{\beta\tilde{E}_\mathbf{k}}{2}\right)$ | Fig. ?? |

Table 1.5 | Symmetry resolved self-consistency equations for the harmonics of $w_\mathbf{k}$.

where $\tau^+ = (\tau^x + i\tau^y)/2$. We used Eqns. (1.33), (1.34) and (1.36) as well as the thermal factor defined in Eq. (1.24). The derivation above holds for all symmetries, (including $s$-wave). Explicitly, five self-consistency equations are then derived and listed in Tab. 1.5.

Of these, the singlet channel admits for non-zero fluctuations of the superconducting order parameter uniquely in the $s^*$ and $d$ channels. This is because the gap function is inversion-symmetric and $p_\ell$ channels equations integrate it with an antisymmetric function.

## 1.7 Full MFT hamiltonian and anisotropic pairing

Let now the total contraction energy

$$E_\mathrm{c}^{(\mathrm{SC})} \equiv -\left[E_{\mathrm{H}/U}^{(\mathrm{SC})} + E_{\mathrm{C}/U}^{(\mathrm{SC})} + E_{\mathrm{H}/V}^{(\mathrm{SC})} + E_{\mathrm{F}/V}^{(\mathrm{SC})} + E_{\mathrm{C}/V/\mathrm{o.s.}}^{(\mathrm{SC})}\right]$$

By composing all three channels we are able to derive the final form of the MFT gran-canonical hamiltonian,

$$\hat{H} - \mu\hat{N} \simeq \sum_{\mathbf{k}\in\mathrm{BZ}} \tilde{\xi}_\mathbf{k} \left[\hat{c}^\dagger_{\mathbf{k}\uparrow}\hat{c}_{\mathbf{k}\uparrow} + \hat{c}_{-\mathbf{k}\downarrow}\hat{c}^\dagger_{-\mathbf{k}\downarrow}\right] - \sum_{\mathbf{k}\in\mathrm{BZ}} \left[\tilde{\Delta}_\mathbf{k}\hat{\phi}_\mathbf{k} + \tilde{\Delta}^*_\mathbf{k}\hat{\phi}^\dagger_\mathbf{k}\right] + E_\mathrm{a}^{(\mathrm{SC})} + E_\mathrm{c}^{(\mathrm{SC})}$$

having we included the "anticommutation energy" also defined for the conventional HM in Eq. (??),

$$E_\mathrm{a}^{(\mathrm{SC})} \equiv \sum_{\mathbf{k}\in\mathrm{BZ}} \tilde{\xi}_\mathbf{k}$$

coming from fermionic anticommutation relations that are employed in order to get the kinetic term. The renormalized bands are

$$\tilde{\xi}_\mathbf{k} = \epsilon_\mathbf{k} + V \sum_{\gamma\in\{s^*,d\}} u^{(\gamma)}\varphi_\mathbf{k}^{(\gamma)*} - \tilde{\mu} \tag{1.55}$$

and the full gap function by

$$\tilde{\Delta}_\mathbf{k} = -Uw^{(s)} + V\sum_\gamma w^{(\gamma)}\varphi_\mathbf{k}^{(\gamma)*} \tag{1.56}$$

We can identify the gap harmonics by

$$\tilde{\Delta}^{(s)} = -Uw^{(s)} \qquad \tilde{\Delta}^{(\gamma)} = Vw^{(\gamma)} \quad \text{being} \quad \gamma \in \{s^*, p_x, p_y, d\}$$

As anticipated, the singlet channel admits three components only,

$$\text{Singlet channel} \quad : \quad \tilde{\Delta}_{\mathbf{k}} = \tilde{\Delta}^{(s)} + \tilde{\Delta}^{(s^*)}(\cos k_x + \cos k_y) + \tilde{\Delta}^{(d)}(\cos k_x - \cos k_y)$$

making the gap function purely real. An inverse statement holds for the triplet pairing, for which

$$\text{Triplet channel} \quad : \quad \tilde{\Delta}_{\mathbf{k}} = i\sqrt{2}\left[\tilde{\Delta}^{(p_x)} \sin k_x + \tilde{\Delta}^{(p_y)} \sin k_y\right]$$

In this case, the gap is purely imaginary. When working in a specific channel, the gap function is either purely real or purely imaginary. This lets us discuss the shape of the renormalised ground-state.

## 1.8 The zero-temperature ground-state at half-filling

The HF approximation to the true ground-state in the SC phase is obtained by aligning the each pseudospin with the respective pseudo-field. Let the up and down states,

$$|\Uparrow_{\mathbf{k}}\rangle \qquad |\Downarrow_{\mathbf{k}}\rangle$$

Using now Eq. (1.23), we have

$$\begin{aligned} \langle \Downarrow_{\mathbf{k}} | \hat{s}^z_{\mathbf{k}} + \hat{\mathbb{I}}_{\mathbf{k}} | \Downarrow_{\mathbf{k}} \rangle &= 0 \\ \langle \Uparrow_{\mathbf{k}} | \hat{s}^z_{\mathbf{k}} + \hat{\mathbb{I}}_{\mathbf{k}} | \Uparrow_{\mathbf{k}} \rangle &= 2 \end{aligned}$$

so up to a phase

$$|\Downarrow_{\mathbf{k}}\rangle \sim |\Omega_{\mathbf{k}}\rangle \qquad |\Uparrow_{\mathbf{k}}\rangle \sim \hat{\phi}^{\dagger}_{\mathbf{k}} |\Omega_{\mathbf{k}}\rangle$$

The derivations carried out for the AF phase in Sec. **??** can be re-used. We end up with the state:

$$|\Psi\rangle = \bigotimes_{\mathbf{k}\in \mathrm{BZ}} \left[\cos\tilde{\theta}_{\mathbf{k}} + e^{2i\tilde{\zeta}_{\mathbf{k}}} \sin\tilde{\theta}_{\mathbf{k}} \hat{\phi}^{\dagger}_{\mathbf{k}}\right] |\Omega_{\mathbf{k}}\rangle$$

which is an extension of the conventional BCS state. The phase factor angle is the gap "tilt" in the complex plane. This is proven by:

$$\begin{aligned} \langle \Psi | \hat{s}^x_{\mathbf{k}} | \Psi \rangle &= \langle \Psi | \hat{\phi}_{\mathbf{k}} + \hat{\phi}^{\dagger}_{\mathbf{k}} | \Psi \rangle \\ &= \sin\tilde{\theta}_{\mathbf{k}} \cos\tilde{\theta}_{\mathbf{k}} \left(e^{2i\tilde{\zeta}_{\mathbf{k}}} + e^{-2i\tilde{\zeta}_{\mathbf{k}}}\right) \\ &= \sin(2\tilde{\theta}_{\mathbf{k}}) \cos(2\tilde{\zeta}_{\mathbf{k}}) \end{aligned}$$

$$\begin{aligned} \langle \Psi | \hat{s}^y_{\mathbf{k}} | \Psi \rangle &= i \langle \Psi | \hat{\phi}_{\mathbf{k}} - \hat{\phi}^{\dagger}_{\mathbf{k}} | \Psi \rangle \\ &= i \sin\tilde{\theta}_{\mathbf{k}} \cos\tilde{\theta}_{\mathbf{k}} \left(e^{-2i\tilde{\zeta}_{\mathbf{k}}} - e^{2i\tilde{\zeta}_{\mathbf{k}}}\right) \\ &= \sin(2\tilde{\theta}_{\mathbf{k}}) \sin(2\tilde{\zeta}_{\mathbf{k}}) \end{aligned}$$

These results are coherent with Eqns. (1.33) as well as (1.34) and allow us to identify the phase with angle of the pseudofield in the $xy$ plane.

## 1.9 SC free energy density

Free energy density of the SC phase is derived following the procedure used for the other two phases. As for the AF phase of Sec. **??**, two fermionic bands are present; however, recalling the antiferromagnet entropy density of Eq. (**??**), the key differences are here given by the fact that the chemical potential inside the Fermi-Dirac factors is null and the sum is extended on the full BZ (but not later on spin index),

$$\begin{aligned} s^{(\pm)} \equiv -\frac{k_B}{L_x L_y} \sum_{\mathbf{k}\in \mathrm{BZ}} \Big[ &\Big(1 - f(\pm\tilde{E}_{\mathbf{k}}; \beta, 0)\Big) \ln\Big(1 - f(\pm\tilde{E}_{\mathbf{k}}; \beta, 0)\Big) \\ &+ f(\pm\tilde{E}_{\mathbf{k}}; \beta, 0) \ln f(\pm\tilde{E}_{\mathbf{k}}; \beta, 0)\Big] \end{aligned} \tag{1.57}$$

with identical meaning of the notation. Using the relation

$$\frac{1}{e^x+1}+\frac{1}{e^{-x}+1}=1$$

we conclude that

$$s^{(-)}=s^{(+)} \qquad \Longrightarrow \qquad s\equiv s^{(-)}+s^{(+)}=2s^{(+)}$$

being $s$ the total entropy density. The five contributions to free energy are:

1. Quasiparticles energy:

$$\begin{aligned} e_{\mathrm{g}}^{(\mathrm{SC})} &= \frac{1}{L_xL_y}\sum_{\mathbf{k}\in\mathrm{BZ}}\tilde{E}_{\mathbf{k}}\tilde{\mathrm{Th}}_{\mathbf{k}}^{(-)}[\beta] \\ &= -\frac{1}{L_xL_y}\sum_{\mathbf{k}\in\mathrm{BZ}}\tilde{E}_{\mathbf{k}}\tanh\left(\frac{\beta\tilde{E}_{\mathbf{k}}}{2}\right) \end{aligned}$$

2. Anticommutation energy:

$$\begin{aligned} e_{\mathrm{a}}^{(\mathrm{SC})} &\equiv \frac{E_{\mathrm{a}}^{(\mathrm{SC})}}{L_xL_y} \\ &= \frac{1}{L_xL_y}\sum_{\mathbf{k}\in\mathrm{BZ}}\tilde{\xi}_{\mathbf{k}} \end{aligned}$$

3. Contraction energy:

$$e_{\mathrm{c}}^{(\mathrm{SC})}=-U\left[n^2+\left|w^{(s)}\right|^2\right]+V\left[2zn^2-\sum_{\gamma}\left|u^{(\gamma)}\right|^2+\sum_{\gamma}\left|w^{(\gamma)}\right|^2\right]$$

4. Entropic energy

$$e_{s}^{(\mathrm{SC})}=\frac{2}{\beta L_xL_y}\sum_{\mathbf{k}\in\mathrm{BZ}}\ln\Big(1-f(\tilde{E}_{\mathbf{k}};\beta,0)\Big)-\frac{2}{L_xL_y}\sum_{\mathbf{k}\in\mathrm{BZ}}\tilde{E}_{\mathbf{k}}f(\tilde{E}_{\mathbf{k}};\beta,0)$$

5. Chemical potential correction

$$e_{n}^{(\mathrm{SC})}=\mu n$$

Compose everything

$$f_{\mathrm{MFT}}^{(\mathrm{SC})}\equiv e_{\mathrm{g}}^{(\mathrm{SC})}+e_{\mathrm{a}}^{(\mathrm{SC})}+e_{\mathrm{c}}^{(\mathrm{SC})}+e_{s}^{(\mathrm{SC})}+e_{n}^{(\mathrm{SC})}$$

and use that:

$$\tanh\left(\frac{x}{2}\right)+\frac{2}{e^x+1}=1$$

In the end we get:

$$\begin{aligned} f_{\mathrm{MFT}}^{(\mathrm{SC})}=\frac{1}{L_xL_y}\sum_{\mathbf{k}\in\mathrm{BZ}}\left[\tilde{\xi}_{\mathbf{k}}-\tilde{E}_{\mathbf{k}}\right]+\frac{2}{\beta L_xL_y}\sum_{\mathbf{k}\in\mathrm{BZ}}\ln\Big(1-f(\tilde{E}_{\mathbf{k}};\beta,0)\Big) \\ -U\left[n^2+\left|w^{(s)}\right|^2\right]+V\left[2zn^2-\sum_{\gamma}\left|u^{(\gamma)}\right|^2+\sum_{\gamma}\left|w^{(\gamma)}\right|^2\right]+\mu \end{aligned}$$

As was for the normal and AF phases, we recognize the chemical potential shift of Eq. (1.45),

$$\begin{aligned} f_{\mathrm{MFT}}^{(\mathrm{SC})}=\frac{1}{L_xL_y}\sum_{\mathbf{k}\in\mathrm{BZ}}\left[\tilde{\xi}_{\mathbf{k}}-\tilde{E}_{\mathbf{k}}\right]+\frac{2}{\beta L_xL_y}\sum_{\mathbf{k}\in\mathrm{BZ}}\ln\Big(1-f(\tilde{E}_{\mathbf{k}};\beta,0)\Big) \\ -U\left|w^{(s)}\right|^2-V\left[\sum_{\gamma}\left|u^{(\gamma)}\right|^2-\sum_{\gamma}\left|w^{(\gamma)}\right|^2\right]+\tilde{\mu}n \end{aligned} \tag{1.58}$$

### 1.9.1 The gap energy

Similarly to Sec. **??**, let us prove the following equation about the gap energy:

$$-U\left|w^{(s)}\right|^2 + V\sum_\gamma \left|w^{(\gamma)}\right|^2 = \frac{1}{2L_xL_y}\sum_{\mathbf{k}\in\mathrm{BZ}} \frac{\left|\tilde{\Delta}_{\mathbf{k}}\right|^2}{\tilde{E}_{\mathbf{k}}}\tanh\left(\frac{\beta\tilde{E}_{\mathbf{k}}}{2}\right) \tag{1.59}$$

Expand the right-hand side through Eq. (1.56),

$$\begin{aligned}\frac{1}{2L_xL_y}\sum_{\mathbf{k}\in\mathrm{BZ}} \frac{\left|\tilde{\Delta}_{\mathbf{k}}\right|^2}{\tilde{E}_{\mathbf{k}}}\tanh\left(\frac{\beta\tilde{E}_{\mathbf{k}}}{2}\right) &= \frac{1}{2L_xL_y}\sum_{\mathbf{k}\in\mathrm{BZ}}\left[-Uw^{(s)*} + V\sum_\gamma w^{(\gamma)*}\varphi_{\mathbf{k}}^{(\gamma)}\right]\frac{\tilde{\Delta}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}}\tanh\left(\frac{\beta\tilde{E}_{\mathbf{k}}}{2}\right)\\ &= -Uw^{(s)*}\times\overbrace{\frac{1}{2L_xL_y}\sum_{\mathbf{k}\in\mathrm{BZ}}\frac{\tilde{\Delta}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}}\tanh\left(\frac{\beta\tilde{E}_{\mathbf{k}}}{2}\right)}^{w^{(s)}}\\ &\quad + V\sum_\gamma w^{(\gamma)*}\times\underbrace{\frac{1}{2L_xL_y}\sum_{\mathbf{k}\in\mathrm{BZ}}\varphi_{\mathbf{k}}^{(\gamma)}\frac{\tilde{\Delta}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}}\tanh\left(\frac{\beta\tilde{E}_{\mathbf{k}}}{2}\right)}_{w^{(\gamma)}}\end{aligned}$$

having we used the self-consistency equations of Tab. 1.5. This proves Eq.(1.59), which is similar to Eq. (**??**) with one exception. Since the right-hand side of Eq.(1.59) is positive, for $V > 0$ we have the constraint

$$\sum_\gamma \left|w^{(\gamma)}\right|^2 \geq M^{(\mathrm{SC})} \quad \text{being} \quad M^{(\mathrm{SC})} \equiv \left|w^{(s)}\right|^2\frac{U}{V} \tag{1.60}$$

Now, as we state in Sec. 1.2.6, for the pure HM we have identically $w^{(s)} = 0$. The pure HM cannot exhibit BCS-like superconductivity. For the EHM instead we see that:

- In the strong coupling limit $U \ll V$, even small $s$-wave gaps impose larger $\gamma \neq s$ components;
- The $s$-wave gap $w^{(s)}$ cannot converge to a finite value if for any $\gamma \neq s$ we have $w^{(\gamma)} = 0$.

Of course, Eq. (1.59) is also solved by a null gap on both sides: this indicates that when the constraint is impossible to fulfil self-consistently, the gap closes.

## 1.10 Discussion of numeric results (singlet SC)

In this section we discuss numeric results for the singlet pairing SC phase. The HFPs vector has five components:

$$\mathbf{v} = \begin{bmatrix} u^{(s^*)} \\ u^{(d)} \\ w^{(s)} \\ w^{(s^*)} \\ w^{(d)} \end{bmatrix} \qquad \mathbf{v} = \mathbb{R}^5$$

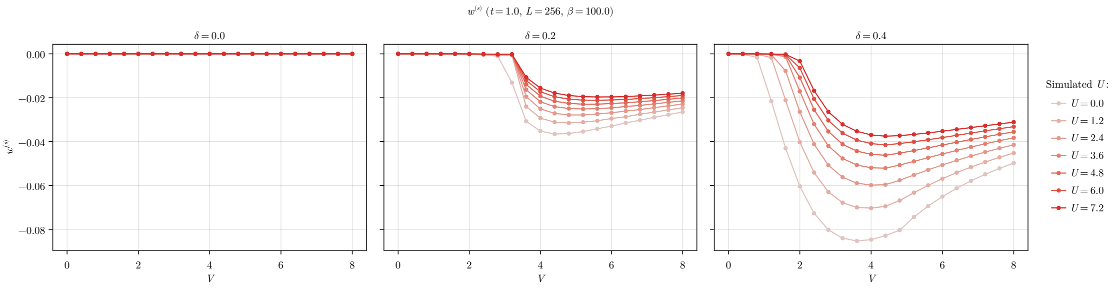
This phase is described by three RMPs:

$$\begin{aligned}\tilde{t}_{ij}^{(s^*)} &\equiv \left(t - \frac{u^{(s^*)}V}{2}\right)\varphi_{ij}^{(s^*)}\\ \tilde{t}_{ij}^{(d)} &\equiv -\frac{u^{(s^*)}V}{2}\varphi_{ij}^{(d)}\\ \tilde{\mu} &\equiv \mu + n(2zV - U)\end{aligned}$$

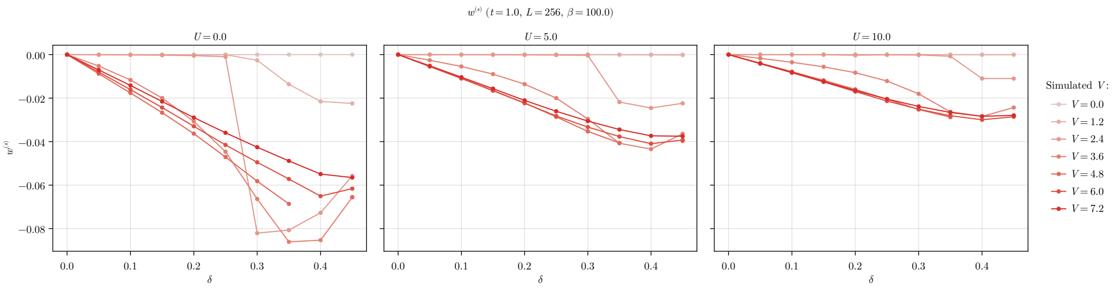
We ran simulations at $\beta = 100.0$ within the boundaries:

$$0 \leq U \leq 8 \qquad 0 \leq V \leq 8 \qquad 0 \leq \delta \leq 0.45$$

For the normal phase we did not vary $U$; for the AF phase we did not vary $\delta$. Here, we need to vary both in order to explore the entire parametric space.




a | Dependency of $w^{(s)}$ on $V$ at various $U$s, compared at three values of $\delta$.




b | Dependency of $w^{(s)}$ on $\delta$ at various $V$s, compared at three values of $U$.

Figure 1.1 | Dependency of the $s$-wave HFP $w^{(s)}$ on $V$ at fixed $U$ and $\delta$ (top panel, compared figures) and on $\delta$ at fixed $V$ and $U$ (bottom panel, compared figures).

### 1.10.1 Multi-symmetry singlet superconductivity

In singlet channel, the most general gap function is a combination of $s$, $s^*$ and $d$ components. The resulting gap function $\tilde{\Delta}_{\mathbf{k}}$ emerges from a competition among these channels. Running unconstrained simulations, we are able to see directly the result of the competition in the self-consistent solution.

We discuss the three channels separately. Note that next figures miss data at $\delta = 0.4$. We were unable to understand why the algorithm failed to converge at that point, and fine-tuning of the mixing parameter $g$ did not help in this case.

**$s$-wave gap.** Let us discuss the local isotropic harmonic of the gap function, namely the $s$-wave component. In Fig. 1.1, we show the data for $w^{(s)}$. The isotropic local gap is given by:

$$\tilde{\Delta}^{(s)} = -Uw^{(s)}$$

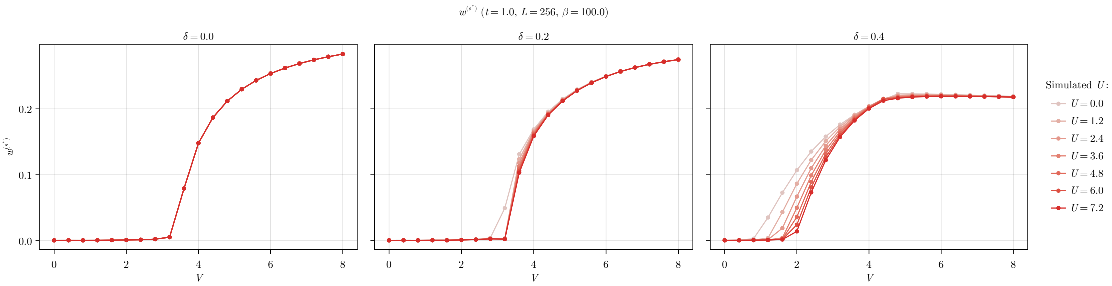
In the conventional HM we know that $s$-wave superconductivity is impossible at HF level. This was discussed in Sec. (1.2.5). Instead in the EHM there is no selection rule for $w^{(s)}$. The only constraint comes from Eq. (1.60): $w^{(s)}$ can be non-zero as long as at least one $w^{(\gamma)}$ with $\gamma = s^*, d$ is non-zero. This is a necessary but insufficient condition.

In Fig. 1.1a we see that the local isotropic gap HFP $w^{(s)}$ is null at $\delta = 0$ (left panel). Hence

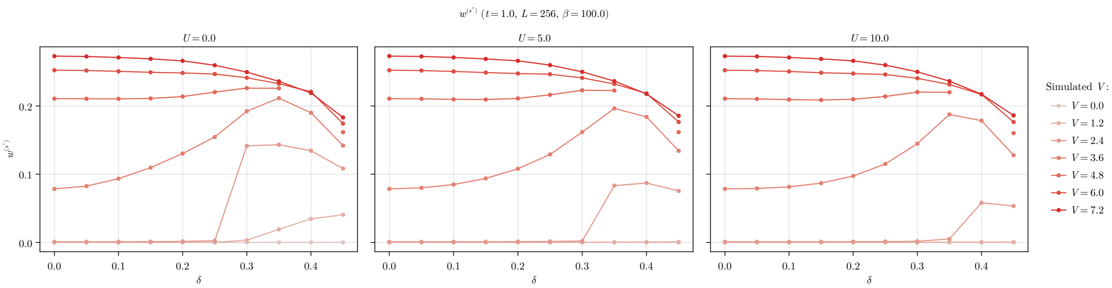
$$\forall\ U/t, V/t \quad : \quad \tilde{\Delta}^{(s)}\big|_{\delta=0} = 0 \tag{1.61}$$

at least within the range of parameters we explored. Increasing doping (central and right panels), $w^{(s)}$ is null up to a critical value $V_c^{(s)}$, which depends also on $U$. For $V \geq V_c^{(s)}$, $w^{(s)}$ is negative and the shape of the curve depends strongly on $U$. In particular, the amplitude $\left|w^{(s)}\right|$ exhibits a local maximum. At almost unitary filling, ratios $V/t$ comparable with 1 are sufficient to develop a finite convergence value for $w^{(s)}$. Therefore, doping lowers the critical value of the non-local attraction that is needed in order to establish an isotropic gap in the local channel.

The dependence of $w^{(s)}$ on doping is shown in Fig. 1.1b. As anticipated, $w^{(s)} = 0$ at half-filling. Varying $U/t$ and $V/t$ we see that $w^{(s)} = 0$ reaches some negative minimum whose amplitude is lower at higher $U$s.




a | Dependency of $w^{(s^*)}$ on $V$ at various $U$s, compared at three values of $\delta$.




b | Dependency of $w^{(s^*)}$ on $\delta$ at various $V$s, compared at three values of $U$.

Figure 1.2 | Dependency of the $s^*$-wave HFP $w^{(s^*)}$ on $V$ at fixed $U$ and $\delta$ (top panel, compared figures) and on $\delta$ at fixed $V$ and $U$ (bottom panel, compared figures).

**$s^*$-wave gap.** The results for the non-local isotropic HFP $w^{(s^*)}$ are shown in Fig. 1.2. This component of the full gap is given by

$$\tilde{\Delta}^{(s^*)} = V w^{(s^*)}$$

Consider first Fig. 1.2a. At null and finite fillings, $w^{(s^*)}$ is null up to a critical value $V_c^{(s^*)}$. Its position depends on $\delta$ and weakly on $U/t$. The half-filling curve (left panel) is independent of $U/t$. Differently from $w^{(s)}$, we have

$$\forall\ U/t \quad \wedge \quad \forall V < V_c^{(s^*)} \quad : \quad \tilde{\Delta}^{(s)}\big|_{\delta=0} = 0$$

The local isotropic gap harmonic $\tilde{\Delta}^{(s)}$ was always null at half filling. Its non-local counterpart $\tilde{\Delta}^{(s^*)}$ is null up to the critical value delimiting the phase transition. Therefore, superconductivity can establish at half-filling with a gap whose isotropic part is purely non-local. Note that we are not saying that pure $s^*$-wave superconductivity is possible. As we discuss later, at half-filling the dominant harmonic of the gap function has $d$-wave symmetry.

In Fig. 1.2b we see that the dependence on doping of $w^{(s^*)}$ is peculiar. At low doping ($\delta \lesssim 0.3$), $w^{(s^*)}$ is small for small values of $V$ (left side of the three panels). At higher doping (right side) we see that a local maximum emerges, indicating that $w^{(s^*)}$ converges to higher values at smaller $V$s.

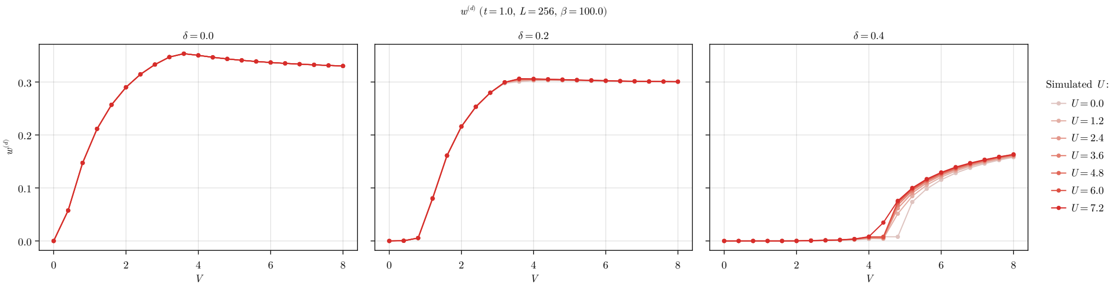
**$d$-wave gap.** In Fig. 1.3 we report our findings for the HFP $w^{(d)}$, recalling that the $d$-wave gap is given by:

$$\tilde{\Delta}^{(d)} = V w^{(d)}$$

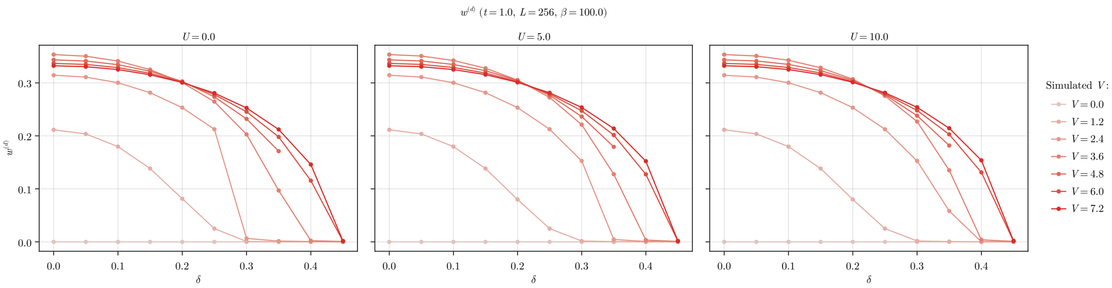
We see from the top panel (Fig. 1.3a) that $w^{(d)}$ depends weakly on $U$. At half-filling (left panel), a $d$-wave gap is present at any $V > 0$:

$$\forall\ U/t \quad \wedge \quad V = 0 \quad : \quad \tilde{\Delta}^{(s)}\big|_{\delta=0} = 0$$

This is a key difference with respect to $w^{(s)}$, $w^{(s^*)}$. The self-consistent superconducting order parameter can be purely $d$-wave, or a mixture of $d$ and $s^*$ symmetries. In the first case, the gap




a | Dependency of $w^{(d)}$ on $V$ at various $U$s, compared at three values of $\delta$.




b | Dependency of $w^{(d)}$ on $\delta$ at various $V$s, compared at three values of $U$.

Figure 1.3 | Dependency of the $d$-wave HFP $w^{(d)}$ on $V$ at fixed $U$ and $\delta$ (top panel, compared figures) and on $\delta$ at fixed $V$ and $U$ (bottom panel, compared figures).

changes sign when subject to rotations of angle $\pi/2$; in the second no symmetry is present under said rotations. At finite filling (central and right panels) a critical value $V_{\mathrm{c}}^{(d)}$ emerges such that the $d$-wave gap appears at $V > V_{\mathrm{c}}^{(d)}$.

The dependence of $w^{(d)}$ on doping at different values of $U$ is shown in the bottom row (Fig. 1.3b). For any $V$, the HFP $w^{(d)}$ is a monotonically decreasing function of $\delta$. As we said, the critical value $V_{\mathrm{c}}^{(d)}$ increases with doping. Because of this, we see that the lighter curves of Fig. 1.3b, corresponding to lower values of $V$, reach zero quite rapidly. The curves for larger values of $V$, instead, reach zero at almost unitary filling. In all three panels of Fig. 1.3b we see that at doping $\delta \approx 0.3$ all lines intersect. In particular, decreasing $V$ the curve gets steeper. At low doping, there exists a finite value of $V$ that maximises $w^{(d)}$.

**Competition among the channels.** To get a valid understanding of the competition among channels, we need to compare the values of the harmonics $\tilde{\Delta}^{(\gamma)}$ for $\gamma = s, s^*, d$. Competition reflects in the relative ratio between different components. We report a comparison of such in Fig. 1.4. The presented data are limited to the case $U = 4.0$ for graphic reasons **,but data for other values of $U$ are openly accessible at our repository.**

On the left column we plot three heatmaps, one for each harmonic in singlet SC phase. The same data are shown, superimposed, in the surface plot on the right. The harmonics $\tilde{\Delta}^{(\gamma)}$ cross if $V/t$ gets large enough, with the $d$-wave component reaching the largest values in the parametric region we investigated. Isotropic harmonics ($s$ and $s^*$) are significantly non-zero in the same region, as we see in the first two rows of the left column. The two channel differ in amplitude significantly: the maximum value reached by the local gap harmonic $\tilde{\Delta}^{(s)}$ is approximately 4 times smaller than the maximum of the non-local harmonic $\tilde{\Delta}^{(s^*)}$. Moreover, we see that at $\delta = 0.0$ the local isotropic harmonic $\tilde{\Delta}^{(s)}$ is completely suppressed, as we expressed in Eq. (1.61).

All harmonics share a quite strange boundary around $\delta \approx 0.34$, $V \approx 2.5$, visible as a sharp vertical segment in the three heatmaps on the left column of Fig. 1.4. The isotropic harmonics suddenly change from zero to large values moving from left to right, while the $d$-wave part does the opposite. Whether this behaviour underlies some deeper structure we were not able to tackle with our coarse-grained simulations, or is a numeric artifact, we do not know.

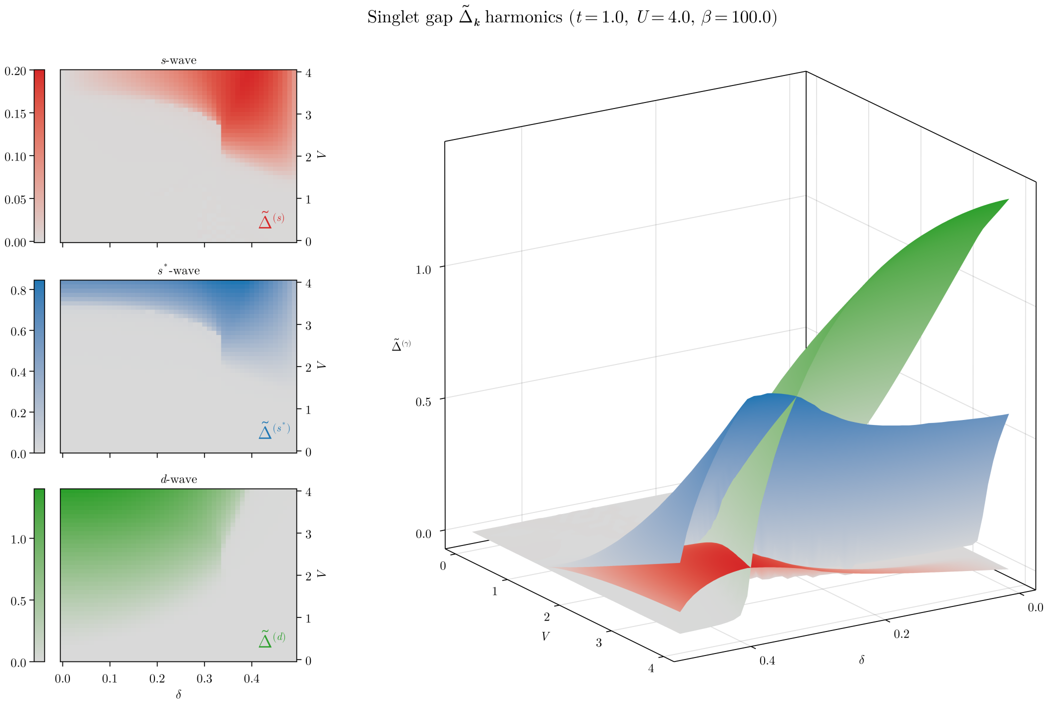



Figure 1.4 Plot of the singlet gap harmonics $\tilde{\Delta}^{(\gamma)}$, with $\gamma = s, s^*, d$ at $U = 4$. The panels on the left show the three harmonics separately as heatmaps, while the surface plot on the right shows them comparatively.

Four phases can be distinguished: the normal state and three superconducting regions (pure $d$-wave, pure $s \oplus s^*$-wave, mixed). In Fig. 1.5 we give a schematic depiction of how the plane $(\delta, V)$ divides in the four regions at $U = 4.0$ (left) and $U = 12.0$ (right). The boundaries were computed by setting an arbitrary threshold $\varepsilon = 10^{-2}$, with the following rule:

$$\begin{array}{rccccc}
\text{Normal} & \left|\tilde{\Delta}^{(s)}\right| < \varepsilon & \wedge & \left|\tilde{\Delta}^{(s^*)}\right| < \varepsilon & \wedge & \left|\tilde{\Delta}^{(d)}\right| < \varepsilon \\
d\text{-wave} & \left|\tilde{\Delta}^{(s)}\right| < \varepsilon & \wedge & \left|\tilde{\Delta}^{(s^*)}\right| < \varepsilon & \wedge & \left|\tilde{\Delta}^{(d)}\right| \geq \varepsilon \\
s \oplus s^*\text{-wave} & \left[\,\left|\tilde{\Delta}^{(s)}\right| \geq \varepsilon\right. & \vee & \left.\left|\tilde{\Delta}^{(s^*)}\right| \geq \varepsilon\,\right] & \wedge & \left|\tilde{\Delta}^{(d)}\right| < \varepsilon \\
\text{Mixed} & \left[\,\left|\tilde{\Delta}^{(s)}\right| \geq \varepsilon\right. & \vee & \left.\left|\tilde{\Delta}^{(s^*)}\right| \geq \varepsilon\,\right] & \wedge & \left|\tilde{\Delta}^{(d)}\right| \geq \varepsilon
\end{array}$$

We consider $s$ and $s^*$ as part of the same isotropic channel: just one harmonic being larger than the threshold is sufficient for us to assert that the gap function has an isotropic harmonic.

Comparing the two panels of Fig. 1.5, we see that increasing the value of the local repulsion $U$ the shapes of the four regions changes. In particular, the $d$-wave region expands to the right (larger $\delta$s); the normal-$s \oplus s^*$ SC boundary moves up (larger $V$s). The mixing region is pretty much unchanged. At low doping, it is bounded from below at $V/t \approx 3$. Increasing doping, we see that the lower boundary reaches a cusp-like minimum around $\delta \approx 0.3$ and then grows. This cusp coincides with the aforementioned sharp vertical segment, visible in Fig. 1.4. If we work at fixed doping and $U$, aligning with the horizontal position of the cusp on the $(\delta, V)$ plane, the critical value of $V/t$ for transitioning to a mixed SC state is minimised.

Symmetry mixing is interesting for another reason: its connection with nematicity. In next section we show that mixing is the root cause of nematic symmetry breaking,

$$\text{Symmetry mixing} \quad \Longrightarrow \quad \mathrm{C}_4 \overset{\text{SSB}}{\to} \mathrm{C}_2$$

The mixed-symmetry region coincides with the portion of the $(\delta, V)$ plane where planar rotational symmetry is broken.

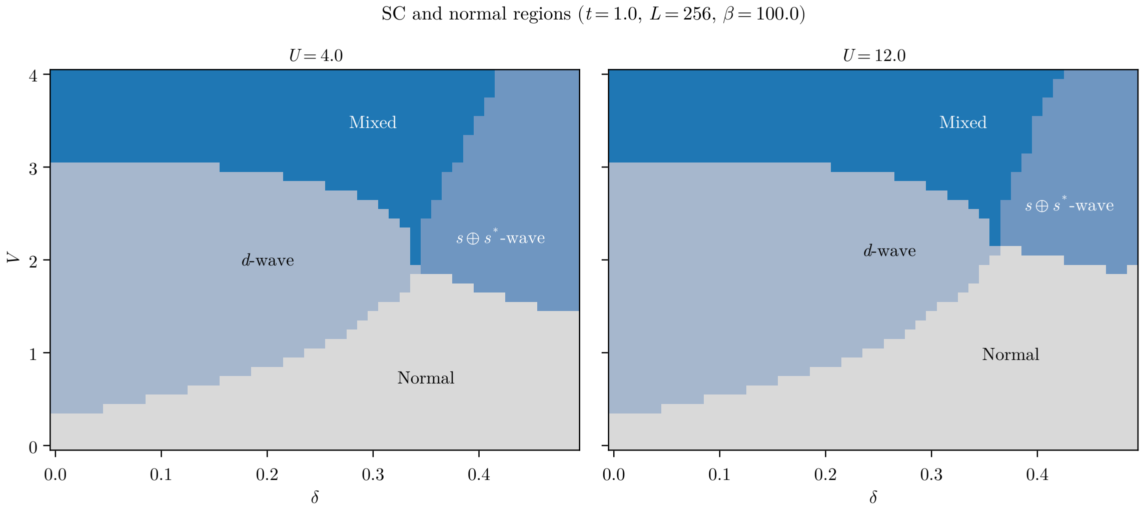



Figure 1.5 Depiction of the normal state region and the three singlet superconducting regions, at $U = 4.0$ (left) and 12.0 (right). Lighter blue indicates the pure $d$-wave SC phase, blue the isotropic $s \oplus s^*$ phase, deeper blue the mixed region. In text is described the convention we used to identify the boundaries.

### 1.10.2 Rotational SSB and nematic superconductivity

In the context of this thesis, nematicity is established whenever $\tilde{t}^{(d)}$ becomes non-zero. This was discussed in detail in the context of the normal phase. If the renormalised hopping parameter has a non-zero $d$-wave component, this means that bare hopping is effectively boosted in one direction and damped in the other. Indeed, from Eq. (**??**), we know that

$$\tilde{t}^{(d)} = \frac{\tilde{t}^{(y)} - \tilde{t}^{(x)}}{2}$$

As we anticipated at the end of Sec. **??**, if $\tilde{t}^{(d)} > 0$ the renormalised hopping is larger in direction $y$, if $\tilde{t}^{(d)} < 0$ instead it's larger in direction $x$. This interpretation serves to grasp the general idea behind nematicity, but is to be taken carefully. Let us get more rigorous. In the SC phase, the many body state is a gas of Bogoliubov quasiparticles which do not "hop on neighbouring sites". If $\tilde{t}^{(d)}$ is non-zero, the superconductor is nematic in the sense that Bogoliubov quasibands are not rotationally invariant,

$$\tilde{E}_{\mathbf{k}} \neq \tilde{E}_{\mathcal{R}\mathbf{k}}$$

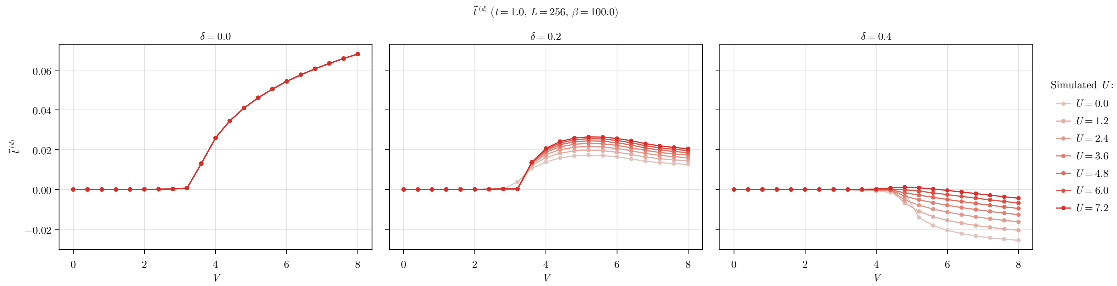
being $\mathcal{R}$ a $\pi/2$ rotation. We connect this effect to hopping renormalisation *a posteriori* following the formal discussion of Sec. **??**.

For the normal and AF phases we were able to prove that $u^{(d)} = 0$ for any choice of model parameters. For the SC phase, instead, $u^{(d)}$ converges to non-zero values. In Fig. 1.6, we plot the convergence value of

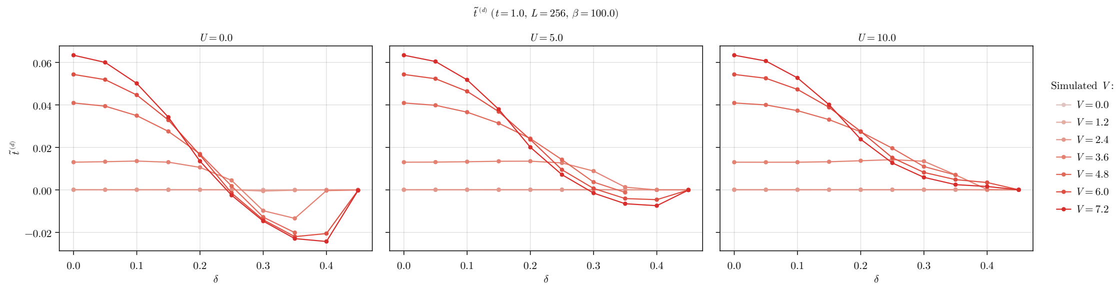
$$\tilde{t}^{(d)} = -\frac{Vu^{(d)}}{2}$$

instead of the HFP $u^{(d)}$: nematicity is undoubtedly present. In the top panel (Fig. 1.6a), we see that a critical value of $V$ exists over which nematicity is established. Moreover, the $\delta = 0.0$ case (left panel) is independent of $U$. In all cases, the convergence values for $\tilde{t}^{(d)}$ are two order of magnitudes smaller than $t = 1.0$: in the region of parametric space we explore, nematicity is a sharp but small effect. Note that at $\delta = 0.2$ (central panel) $\tilde{t}^{(d)} \geq 0$, while at $\delta = 0.4$ (right panel) $\tilde{t}^{(d)} \leq 0$. Increasing the doping we change the sign of $\tilde{t}^{(d)}$

This sign change is most evident in the first two figures of the bottom panel (Fig. 1.6b), where we vary the doping level while keeping fixed the local repulsion $U$. The behaviour of $\tilde{t}^{(d)}$ is similar at low doping in the three cases, but changes significantly at larger doping. Indeed, at $U = 0.0$ (left figure) and $U = 5.0$ (central figure) at some point $\tilde{t}^{(d)}$ changes sign. At $U = 10.0$, all our




a | Dependency of $t^{(d)}$ on $V$ at various $U$s, compared at three values of $\delta$.




b | Dependency of $t^{(d)}$ on $\delta$ at various $V$s, compared at three values of $U$.

Figure 1.6 | Dependency of the $d$-wave renormalised hopping $t^{(d)}$ varying the doping $\delta$ and the local repulsion $U$. In the top panel we compare three doping levels $\delta = 0.0,\ 0.2,\ 0.4$ varying $0 \leq U \leq 8$. In the bottom panel we compare three repulsion values $U = 0.0,\ 5.0,\ 10.0$ varying $0 \leq \delta \leq 0.45$.

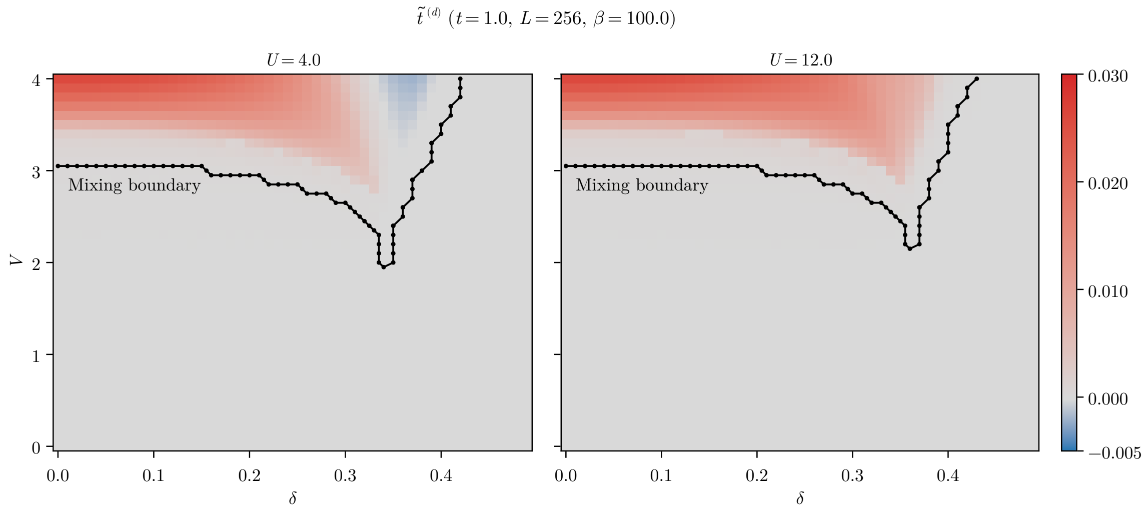
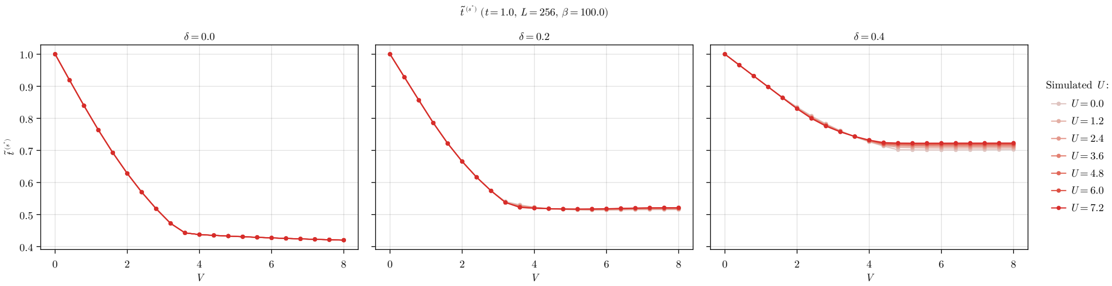


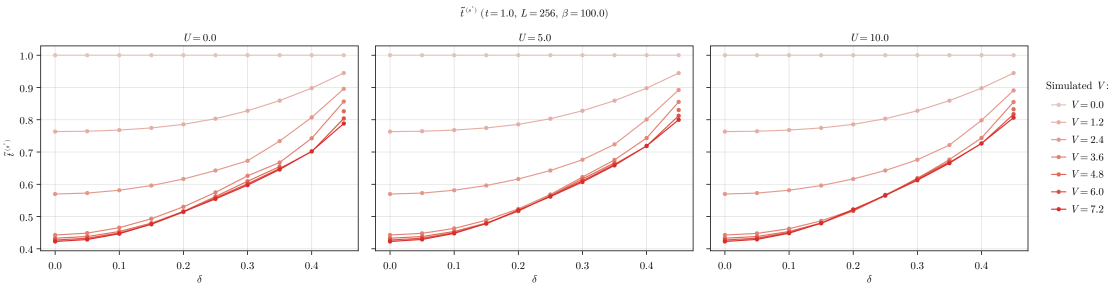

Figure 1.7 | Heatmap of the anisotropic component of renormalised hopping, $\tilde{t}^{(d)}$, at $U = 4.0$ (left) and 12.0 (right). The dataset is the same of Fig. 1.5, and the mixing boundary is the set of points separating the mixed region from the rest of the phases.

data points are positive (possibly the sign flip occurs at higher $V$s). Moreover, as $U$ increases, the amplitude of the oscillation of $\tilde{t}^{(d)}$ as a function of $\delta$ reduces. We conclude that both the local repulsion and electronic doping disrupt nematicity, which is instead most effective for a purely NNs attractive model.

Nematic effects are directly related to symmetry mixing in the superconducting order parameter. Consider Fig. 1.7, where we plot the convergence values for $\tilde{t}^{(d)}$ in the $(\delta, V)$ plane at $U = 4.0, 12.0$. The used dataset is the same of Fig. 1.5. The superimposed black lines are the




a | Dependency of $t^{(s)}$ on $V$ at various $U$s, compared at three values of $\delta$.




b | Dependency of $t^{(s)}$ on $\delta$ at various $V$s, compared at three values of $U$.

Figure 1.8 | Dependency of the $s^*$-wave renormalised hopping $t^{(s^*)}$ varying the doping $\delta$ and the local repulsion $U$. In the top panel we compare three doping levels $\delta = 0.0,\ 0.2,\ 0.4$ varying $0 \leq U \leq 8$. In the bottom panel we compare three repulsion values $U = 0.0,\ 5.0,\ 10.0$ varying $0 \leq \delta \leq 0.45$.

mixing boundaries, the set of extremal points of the deeper blue regions of Fig. 1.5. Despite a bit of roughness related to the gridded nature of our data, it is evident that these boundaries encircle the regions where $\tilde{t}^{(d)} \neq 0$. Note that the set of points composing the mixing regions are those satisfying the threshold rule exposed in last section; choosing another threshold, the curves change. Nevertheless, the key point is that there is an evident connection between symmetry mixing and breaking of planar rotational symmetry.

The two panels of Fig. 1.7 differ crucially in the fact that the data for $U = 12.0$ (right) are all positive, while those for $U = 4.0$ (left) are negative in a small portion of space. With a small abuse of notation, we can identify:

$$\mathrm{M} \simeq \mathrm{N}_x \cup \mathrm{N}_y$$

with the following convention on notation:

$$\begin{aligned} \mathrm{M} &= \text{Region of } (\delta, V) \text{ plane where mixing occurs} \\ \mathrm{N}_x &= \text{Region of } (\delta, V) \text{ plane where } \tilde{t}^{(d)} < 0 \\ \mathrm{N}_y &= \text{Region of } (\delta, V) \text{ plane where } \tilde{t}^{(d)} > 0 \end{aligned}$$

The mixing region divides in two parts, based on the sign of $\tilde{t}^{(d)}$. In $\mathrm{N}_\ell$, the index $\ell$ indicates the direction along which renormalised hopping is larger.

Note that the map were colour-corrected in order to enhance the intensity of the blue region

### 1.10.3 Isotropic bands shrinking

Breaking of planar rotational symmetry and the insurgence of a $d$-wave harmonic for the renormalised hopping is certainly the most peculiar effect for the singlet SC phase. As was for the AF and normal phases, however, Mott localisation is present also here. This effect impacts the $s^*$-wave harmonic of the renormalised hopping,

$$\tilde{t}^{(s^*)} = t - \frac{V u^{(s^*)}}{2}$$

which is shown in the panels of Fig. 1.8. As was for the $d$-wave harmonic, we plot directly the data for $\tilde{t}^{(s^*)}$.

In the top panel (Fig. 1.8a), the dependence of $t^{(s^*)}$ on $V$ is in general pretty similar to AF phase.

### 1.10.4 Solution free energy and entropy densities

[To be continued...]

## 1.11 A final note

[To be continued...]

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